Trees, Forests and People 26 (2026) 101395

Contents lists available at ScienceDirect

Trees, Forests and People

journal homepage: www.sciencedirect.com/journal/trees-forests-and-people

Highway edge spaces as urban green infrastructure corridors: A typology-based NBS suitability framework for rapidly urbanising cities in the Global South

Gokul Kailash *, Banu Chitra M

School of Planning and Architecture, Vijaywada, Andhra Pradesh, 520008, India

ARTICLE INFO

Keywords:
Highway edge spaces
Nature-based solutions
Spatial decision-support
Urban green infrastructure
GIS
Chennai
Global South
Residual urban spaces

ABSTRACT

Highway edge spaces narrow, linearly distributed residual strips generated by road infrastructure expansion represent a systematically overlooked typology for urban green infrastructure establishment in rapidly urbanising cities of the Global South. Despite their extensive spatial coverage along arterial corridors, these infrastructure-induced margins remain excluded from formal vegetation planning, perpetuating ecological degradation, habitat fragmentation, and green space inequity in adjacent high-density neighbourhoods. This study develops a typology-based Nature-Based Solutions (NBS) suitability framework for the identification, classification, and prioritisation of highway edge spaces as urban green infrastructure corridors across the Greater Chennai Corporation (GCC) area, India. Highway edge spaces are mapped and classified into distinct morphological typologies grade-level margins, elevated flyover understories, and median strips each assessed against measurable site-level criteria including edge width, surface condition, vertical clearance, and pedestrian accessibility. Green space deficit is quantified at ward level, revealing a severe vegetation deficit: 133 of 135 GCC wards fall below 1 m^2 of green space per capita, all 135 wards fall below the UN-recommended threshold of 9 m^2, and median green cover stands at just 0.62% of ward area, with the most critically underserved wards recording per capita provision as low as 0.0005 m^2. A socio-ecological accessibility layer integrating Point of Interest density, population density, and walkability contextualises green infrastructure priority within equity conditions of adjacent communities. Spatial overlay of these three analytical layers identifies 82 Priority Intervention Corridors approximately 41% of GCC wards concentrated along Chennai's five major highway alignments, where NBS deployment can simultaneously restore urban vegetation cover, improve ecological connectivity, and extend green infrastructure access to underserved populations. The framework offers a replicable spatial planning instrument for Global South cities confronting critical deficits in accessible urban green space.

1. Introduction

Rapid urbanisation in cities across the Global South has produced a well-documented but poorly resolved planning problem: the generation of highway edge spaces. As road networks are widened, elevated, and extended to serve growing urban populations, narrow strips of land are left along corridor margins fragmented, environmentally degraded, excluded from formal green infrastructure investment, and located in the immediate vicinity of densely populated, underserved neighbourhoods. These spaces are not incidental. Traditional planning practices, wherein attention is directed at the provision of single functions or zoning, have led to the emergence of lost spaces in cities, and elevated highways are identified as prominent contributors to the formation of such spaces (Anuar and Abdullah, 2020;). Their formation is structural, repeated across cities wherever infrastructure expansion has outpaced the planning capacity to manage what is left behind.

The consequences of this condition are significant for urban greenness and urban liveliness alike. Global South cities experience more rapid urban green space decline than those in the Global North, with urbanisation identified as the dominant force driving this loss, and context-specific urban greening strategies are considered crucial for equitable and sustainable cities. In Indian cities, this pressure is particularly acute. Increase in population has adversely affected green cover in urban India, with Chennai recording green space per capita as low as

* Corresponding author.
E-mail address: ar.gokulkailash@gmail.com (G. Kailash).

https://doi.org/10.1016/j.tfp.2026.101395

Available online 8 July 2026
2666-7193/© 2026 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

0.46 m^2, far below the UN-recommended standard of 9 m^2 (Govindarajulu, 2014). Communities settled alongside highway corridors bear the environmental burden of road infrastructure while remaining among the least served by accessible green space a condition in which greenness deficit and urban liveliness deficit are spatially concentrated in the same neighbourhoods.

Despite growing recognition of the ecological and social potential of underutilised urban land, a replicable planning framework for converting highway edge spaces into NBS-led regeneration corridors has not been established. In rapidly urbanising cities where formal parks and green networks are costly and difficult to implement, underutilised spaces represent a scalable and flexible alternative for embedding green infrastructure into dense or informally developed areas, yet most existing approaches have failed to provide a coherent framework with multiple options for converting such spaces into green infrastructure, with the majority of studies limited to site identification and classification. This gap is particularly pronounced in Global South cities, where institutional constraints, ambiguous land tenure, and fragmented governance compound the challenge of reclaiming marginal infrastructure land for public use.

Nature-Based Solutions are positioned in this paper as the appropriate instrument for addressing that gap. A growing number of cities worldwide are exploring NBS, which incorporate natural processes to support adaptation to climate change and urbanisation as well as the restoration of ecological functions, and are promoted for their co-benefits to individuals and communities including improved public health, access to nature, and social equity. When applied within linear infrastructure corridors, NBS carry the potential to simultaneously restore ecological function, reduce urban heat, manage stormwater, improve environmental quality, and generate the spatial conditions associated with urban liveliness understood here as the intensity of human activity, social engagement, and functional diversity in public space.

A GIS-based NBS framework is developed and demonstrated in this paper for the reclamation of highway edge spaces as greenness-led urban regeneration corridors. Chennai, India is selected as the demonstrative case. Per capita provision of urban green spaces in Chennai stands at less than 1 sq. m., far below the norms of 10–12 sq. m. recommended by the Government of India (Govindarajulu, 2014), and decades of highway expansion across the Greater Chennai Corporation area have generated extensive edge spaces along arterial corridors that remain outside formal planning. The conditions present in Chennai infrastructure-induced residual land, acute green space deficit, and inequitable distribution of urban liveliness are shared across a wide range of rapidly urbanising cities. The framework developed here is therefore presented as a transferable planning instrument, with Chennai serving as the empirical ground through which its applicability and replicability are demonstrated.

Three research questions are addressed. First, where are highway edge spaces located across Chennai and what are their spatial characteristics and NBS suitability? Second, how does NBS suitability correspond spatially with areas of urban liveliness deficit? Third, what does the resulting framework offer as a replicable model for other cities facing comparable pressures? The paper is structured as follows. Section 2 reviews relevant literature on highway edge spaces, NBS, and urban liveliness. Section 3 describes Chennai as the study area and justifies its selection as a representative demonstrative case. Section 4 presents the methodology. Section 5 reports the results. Section 6 discusses the findings against the special issue themes of greenness and city liveliness in urban regeneration, with explicit attention to transferability. Section 7 concludes with planning implications and directions for future research.

2. Literature review

2.1. Highway edge spaces as urban planning problem

Highway edge spaces have been examined in planning literature through several related but distinct framings as lost space, as residual land, and more recently as underutilised urban resource. The foundational conceptualisation of such spaces as a planning problem is traced to Trancik (1986), who identified the spaces left behind by infrastructure as zones of disconnection that weaken urban fabric and reduce the conditions for public life. This framing has since been extended by scholars examining the specific condition of elevated and grade-level highways. The construction of urban highways has been found to result in a significant amount of residual spaces that are rarely incorporated into official planning and design efforts, with the spaces underneath elevated structures shown to trigger issues ranging from debris dumping to crime (Anuar and Abdullah, 2020;). Such residual spaces have been connected to the incivilities theory, in which abandoned spaces are found to promote weak social ties among nearby residents and encourage antisocial behaviour, with elevated highways described as infrastructure that seldom attracts people's affection and poses a constant provocation to surrounding communities(Harnack and Kohler, 2011).

The ecological dimension of highway edge spaces has been equally well-documented. Road infrastructure is recognised as one of the most significant drivers of habitat fragmentation in urban landscapes, reducing ecological continuity, limiting vegetation density in adjacent zones, and suppressing the conditions for biodiversity in corridor margins (Forman et al., 2003). Studies of transport-adjacent land have consistently found that the mono-functional orientation of highway corridors designed exclusively for vehicular mobility produces edge territories with low ecological performance, poor environmental quality, and minimal social value (Hamersma et al., 2017). Despite this, these same characteristics linearity, adjacency to dense residential areas, and public land ownership are increasingly recognised as assets when viewed through a regenerative lens. Underutilised urban spaces characterised by vacancy, neglect, or fragmented use represent critical yet underexploited opportunities for advancing urban sustainability and resilience through green infrastructure, and in rapidly urbanising cities where formal parks and green networks are costly, such spaces represent a scalable and flexible alternative (Aghaloo and Sharifi, 2024).

Progress in reclaiming infrastructure-adjacent land has, however, been uneven.Case studies from Kuala Lumpur, Colombo, and Sao Paulo have demonstrated site-specific approaches to greening elevated highway residuals, yet these efforts have largely remained at the level of typological classification and site observation without advancing a replicable spatial planning framework (Sanches and Pellegrino, 2016; Anuar and Abdullah, 2020). Analysis across these studies reveals that most approaches fail to provide a coherent framework with multiple options for converting underutilised urban spaces into green infrastructure, with the majority of existing work limited to site identification and classification and very few studies presenting any indications as to how action should be taken.This gap is the methodological departure point for the framework developed in this paper.

2.2. Nature-based solutions: functions, co-benefits, and spatial application

Nature-Based Solutions have emerged over the past decade as a widely adopted conceptual and operational framework for addressing the compounding environmental and social pressures on urban systems. NBS result in the provision of co-benefits such as the improvement of place attractiveness, health and quality of life, and the creation of green jobs, yet few frameworks exist for acknowledging and assessing the value of such co-benefits or for guiding cross-sectoral project and policy design and implementation (Raymond et al., 2017). Their

multifunctionality spanning ecological restoration, stormwater management, urban cooling, biodiversity support, and improvements to public realm quality makes them particularly suited to constrained urban sites where a single intervention must deliver multiple returns (Kabisch et al., 2017). Equity and justice have been identified as key concerns intersecting all domains of NBS implementation, with inclusive approaches required to ensure that marginalised communities are not left behind in the design, planning, deployment, and distribution of NBS benefits (McPhearson et al., 2025). This equity dimension is especially relevant in the context of highway edge spaces, where the communities most exposed to road infrastructure impacts noise, air pollution, physical severance are frequently those with the least access to compensatory green space. When NBS are deployed in highway edge corridors, they thus serve a dual equity purpose: restoring greenness to the immediate environment while directing environmental benefits toward underserved populations.

Despite increasing academic interest, integrating Nature-Based Solutions (NBS) into urban green infrastructure faces challenges due to fragmented methodologies and a limited understanding of how they function in specific local contexts (Chatzimentor et al., 2020). In cities in the Global South, most NBS frameworks stem from Global North experiences, which restricts their effectiveness in settings marked by informality, institutional fragmentation, and limited governance capacity (McPhearson et al., 2025). This paper identifies a GIS-based approach for assessing NBS suitability as a suitable method to address this issue, as it makes spatial decisions explicit, easy to replicate, and adaptable to various urban environments.

2.3. Urban liveliness: definitions, determinants, and the role of greenness

Urban liveliness, often used interchangeably with urban vitality in this paper, refers to the intensity and variety of human activity, movement, and social engagement in urban public spaces over time. Vitality is defined as the degree to which a space feels alive or lively and specifically pertains to how well a space can consistently host diverse activities by heterogeneous populations (Kamalipour and Peimani, 2015; Jacobs, 1961). This foundational concept has been operationalised in modern research through measurable spatial indicators. Urban vitality encompasses aspects such as different paces of pedestrian flow, use of urban infrastructure, and cultural events throughout the year that set successful urban areas apart from others, with vibrancy reflected by human presence and activities (Montgomery, 1998; Jacobs, 1961). Urban vitality is linked to features of public spaces like green areas, water features, mixed-use zones, and pedestrian accessibility (Huang et al., 2024).

The relationship between greenness and urban liveliness is empirically established but varies depending on the context. Urban green spaces can enhance urban vitality and provide quality of life benefits, and there is increasing interest in NBS as a potentially effective solution for various societal challenges that affect both social and environmental aspects simultaneously. Exposure to green space has been consistently associated in the literature with reduced stress, better mental health, and increased social cohesion all of which are seen as key components of urban vitality (van den Berg et al., 2015). However, the relationship is not unconditional: the vitality benefits of green space are heavily influenced by factors such as accessibility, safety, maintenance, and the extent to which these areas support social interaction rather than merely serving as passive amenities (Gómez-Varo et al., 2026).

In infrastructure-adjacent spaces, the connection between greenness and liveliness has not been extensively studied. Highway corridors often experience a simultaneous decrease in greenness and urban liveliness, which are spatially linked. When nature-based solutions (NBS) are applied to highway edge land to restore greenery, this also tends to enhance urban liveliness by encouraging more pedestrian activity, improving environmental perception, and increasing access to green spaces. These improvements are predictable and measurable benefits that co-occur. This idea is central to the Liveliness Potential Layer described in the methodology section.

2.4. Identifying the research gap

This review highlights three gaps in the current literature. First, highway edge spaces are rarely treated as a distinct spatial type in NBS and green infrastructure planning, especially in cities of the Global South. Second, there is no established GIS-based framework for evaluating NBS suitability across different highway edge types at the city level. Third, the impact of NBS-induced greenness in infrastructure corridors on urban vitality has not been spatially analysed or operationalised in dense South Asian cities. The framework proposed in this study aims to fill all three gaps, using Chennai as a case example to demonstrate its applicability and transferability Table 1.

3. Study area

3.1. Geographic scope and administrative boundary

The study focuses on the administrative area of the Greater Chennai Corporation (GCC), which is the main municipal authority for urban management in Chennai city. This jurisdiction spans 426 sq. km., a boundary that has stayed the same since the 2011 expansion, when 42 local bodies were incorporated into the corporation. The population within this boundary is approximately 7.95 million, based on 2023 projections from the 2011 census, with the highest population densities in the central and northern zones. The study area was chosen over the larger Chennai Metropolitan Area (CMA) because the GCC covers the densely developed urban core, where issues like highway infrastructure, residential density, green space shortages, and urban vibrancy are most prominent and comparable across different corridors.

3.2. Urban character and development context

Chennai is the fourth largest metropolis in India and one of the country's most important urban centres economically, with a diverse economic base that includes automobile manufacturing, information technology, healthcare, and port-based trade (CMDA, 2008). Urbanisation has progressed rapidly over the decades, with the city's gross population density recorded at 247 persons per hectare in 2001 and expected to increase significantly during the current planning period (CMDA, 2008). Water bodies in the city have diminished from an estimated 12.6 sq. km. in 1893 to 3.2 sq. km. in 2017, and the number of wetlands has decreased from 650 in 1970 to 27 in 2015, with nearly half of the native plant species in city wetlands disappearing. This ongoing loss of ecological assets due to urbanisation directly supports the study's argument that residual spaces along highway edges are among the few remaining opportunities for integrated green infrastructure within the built-up city.

Per capita provision of urban green spaces in Chennai is less than 1 sq. m., significantly below the 10–12 sq. m. standards recommended by the Government of India (Govindarajulu, 2014). The distribution of this shortfall is uneven: access to green space is most limited in high-density residential areas near major highway corridors, where road infrastructure dominates land use and open spaces are systematically excluded from planning. This spatial overlap of green space deficiency with infrastructure-adjacent populations makes Chennai a pertinent and suitable case for demonstrating the framework developed in this paper.

3.3. Highway infrastructure and edge space generation

Chennai's road network spans approximately 1,780 km as of 2023, with the city connected to the Golden Quadrilateral National Highway system and to Mumbai and Kolkata via NH 48 and NH 16, respectively. Within the GCC boundary, the major highway corridors that form the

Table 1
Summary of key literature reviewed across four thematic areas: highway edge spaces, nature-based solutions, urban liveliness, and GIS spatial methods identifying the conceptual and methodological foundations of the study and the research gaps addressed.

Author(s) & Year	Thematic Area	Key Contribution	Relevance to This Paper
2.1 Highway Edge Spaces as Urban Planning Problem			
Trancik (1986)	Lost spaces	Foundational identification of infrastructure-generated spaces as zones of disconnection that weaken urban fabric and reduce conditions for public life.	Establishes conceptual basis for treating highway edge spaces as a planning problem requiring active intervention.
Harnack and Kohler (2011)	Residual infrastructure land	Connected elevated highway residuals to incivilities theory; abandoned highway edges found to promote weak social ties and encourage antisocial behaviour among adjacent communities.	Underpins the social argument for reclaiming highway edge spaces as active public environments.
Forman et al. (2003)	Road ecology	Road infrastructure identified as a major driver of habitat fragmentation, reducing ecological continuity and suppressing biodiversity in corridor margins.	Justifies NBS as the ecological restoration instrument for highway edge corridors.
Hamersma et al. (2017)	Highway nuisance perception	Mono-functional orientation of highway corridors found to produce edge territories with low ecological performance, poor environmental quality, and minimal social value.	Documents the compounded social–ecological cost of unreclaimed highway edge spaces.
Mohamed Anuar and Abdullah (2020)	Elevated highway residuals	Identified two typologies of residual spaces (easily accessible; hard to access) and assessed green infrastructure suitability for each in Kuala Lumpur; confirmed planning neglect of elevated highway margins.	Key comparative case; typology approach informs classification methodology used in this paper.
Sanches and Pellegrino (2016)	Derelict & vacant urban land	Examined greening potential of derelict and vacant land in São Paulo; confirmed spatial opportunity but noted absence of replicable GI frameworks.	Establishes the Global South gap in actionable GI frameworks for residual land.
Weththasinghe and Wijesundara, 2017	Traffic-influenced residual spaces	Assessed possibilities for reclaiming traffic-influenced urban residual spaces in Colombo; found most efforts limited to site observation without operational planning framework.	Reinforces methodological gap this paper addresses — absence of a city-scale replicable framework.
2.2 Nature-Based Solutions: Functions, Co-benefits, and Spatial Application			
Raymond et al. (2017)	NBS co-benefits framework	Developed a holistic framework for assessing co-benefits of NBS across socio-cultural, biodiversity, and climate dimensions; demonstrated NBS deliver improvement in place attractiveness, health, and quality of life.	Core NBS framework adopted in this paper; co-benefits framing used to link NBS suitability with liveliness potential.
Kabisch et al. (2017)	NBS multifunctionality	Demonstrated that urban NBS deliver multiple simultaneous returns — stormwater management, urban cooling, biodiversity support, and public realm improvement — making them suited to constrained urban sites.	Justifies NBS as the preferred intervention type for narrow, multi-pressure highway edge corridors.
Grabowski et al. (2023)	NBS for urban sustainability	Established that NBS are at the forefront of addressing urban climate impacts and incorporate natural processes to support both adaptation and ecological restoration.	Positions NBS within the urban sustainability agenda and supports the paper's intervention framing.
Frantzeskaki (2019)	NBS implementation in cities	Assessed governance conditions, institutional capacities, and barriers to urban NBS adoption; found urban NBS require alignment between planning instruments and spatial strategy.	Informs governance discussion in Section 6; highlights need for framework portability across institutional contexts.
Chatzimentor et al. (2020)	NBS–GI integration barriers	Systematic review confirmed that practical integration of NBS into urban green infrastructure remains hindered by fragmented methodologies and limited context-specific implementation guidance.	Directly identifies the methodological gap filled by the GIS-based NBS suitability framework developed here.
Lahoti et al., 2019	GIS-based NBS suitability (Nagpur)	Applied GIS spatial analysis to assess urban green space gaps in an Indian city; developed methodology for spatially defining priority areas for green infrastructure establishment.	Closest Indian methodological precedent; suitability approach adapted for highway edge typology in this paper.
McPhearson et al. (2025)	NBS equity & Global South	Identified equity and justice as key concerns in NBS implementation; found Global South cities under-represented in NBS frameworks; local and regional contexts identified as critical to effective NBS implementation.	Justifies the Global South orientation and equity lens applied to Chennai as demonstrative case.
Govindarajulu (2014)	Blue-green infrastructure, Chennai	Documented the acute green space deficit in Chennai and the need for blue-green infrastructure solutions suited to the local South Indian urban context.	Provides Chennai-specific empirical grounding for the NBS argument and supports study area selection.
Raymond et al. (2017)	NBS co-benefits framework	Developed a holistic framework for assessing co-benefits of NBS across socio-cultural, biodiversity, and climate dimensions; demonstrated NBS deliver improvement in place attractiveness, health, and quality of life.	Core NBS framework adopted in this paper; co-benefits framing used to link NBS suitability with liveliness potential.
2.3 Urban Liveliness: Definitions, Determinants, and the Role of Greenness			
Jacobs (1961)	Urban vitality theory	Foundational definition of urban vitality as the level at which a space can consistently accommodate diverse activities by heterogeneous populations; identified mixed use, small blocks, concentration of people, and activity diversity as conditions for liveliness.	Provides the theoretical foundation for the urban liveliness framework applied in this paper.
Gehl (2011)	Human-scale urban design	Documented the relationship between built environment design and the quality and quantity of outdoor human activities; demonstrated that physical space quality directly conditions urban liveliness.	Supports the argument that NBS-activated highway edge spaces can improve the physical conditions for liveliness.
Montgomery (1998)	Public space vitality	Defined vitality as the degree to which a space is alive or lively; identified green areas, water, mixed use, and pedestrian accessibility as public space features associated with urban vitality.	Directly links greenness to vitality — core causal claim of this paper.
Gómez-Varo et al. (2026)	Operationalising Jacobs (Barcelona)	Developed the JANE Index to quantify urban vitality potential using Jacobs' criteria; demonstrated replicability of vitality assessment across different urban contexts.	Provides a replicable vitality quantification approach that informs the Liveliness Potential Layer developed here.
Ye et al. (2018)	Urban vitality measurement	Measured urban vitality using big data and POI density; confirmed functional diversity and pedestrian accessibility as primary quantifiable determinants of liveliness.	POI-based vitality measurement approach referenced in the methodology for the Liveliness Potential Layer.
Xue et al. (2025)	Greenness–liveliness relationship	Demonstrated empirical relationship between urban green space access and measures of urban liveliness, social activity, and mental wellbeing; showed spatial co-location of greenness deficit and vitality deficit.	Empirically grounds the paper's central argument linking NBS suitability with liveliness deficit.

(*continued on next page*)

Table 1 (*continued*)

2.3 Urban Liveliness: Definitions, Determinants, and the Role of Greenness			
Iungman et al. (2023)	Green space and public health	Documented the relationship between urban green space access and reduced heat, improved air quality, and improved mental health outcomes in European cities; highlighted equity dimension of green space distribution.	Supports the co-benefit argument linking NBS activation with improved health and vitality outcomes for highway-adjacent communities.
Jacobs (1961)	Urban vitality theory	Foundational definition of urban vitality as the level at which a space can consistently accommodate diverse activities by heterogeneous populations; identified mixed use, small blocks, concentration of people, and activity diversity as conditions for liveliness.	Provides the theoretical foundation for the urban liveliness framework applied in this paper.
2.4 GIS Spatial Methods & Urban Green Infrastructure Planning			
Peiris (2024)	GIS urban resilience assessment	Developed a GIS-based spatial assessment approach for urban resilience that combines multi-criteria spatial layers; demonstrated replicability across different city types.	Informs the multi-layer GIS methodology applied in this paper's NBS suitability and liveliness assessments.
Sathyakumar et al. (2020)	Urban green space GIS (India)	Applied GIS to assess green space distribution and identify spatial gaps in Indian urban contexts; demonstrated the value of spatial prioritisation for equitable GI planning.	Indian urban GIS precedent; spatial gap analysis approach adapted for the highway edge typology framework.
Lahoti et al. (2019)	Infrastructure corridor typology	Developed a typology for green infrastructure along urban infrastructure corridors; demonstrated that corridor type, land area, and adjacency conditions determine appropriate NBS strategies.	Most directly relevant methodological precedent for the corridor typology and NBS strategy matching applied in this paper.
Govindarajulu (2014)	Urban green space, India	Documented the acute green space deficit across Indian cities including Chennai; estimated Chennai's per capita green space at below 1 sq. m. against a standard of 10–12 sq. m.	Provides the empirical baseline for Chennai's green space deficit against which NBS potential is assessed.
Turaga et al. (2020)	Urban environment, South India	Assessed urban environmental conditions and green space inequity in South Indian cities; documented spatial concentration of environmental deficits in lower-income, infrastructure-adjacent neighbourhoods.	Supports the equity framing of the paper and the spatial co-location argument for highway edge NBS intervention.

spatial backbone of this study are the Inner Ring Road (IRR), the Old Mahabalipuram Road (OMR / Rajiv Gandhi Salai), the Grand Southern Trunk Road (GST Road), the Poonamallee High Road, and Anna Salai (Fig. 1). In addition, an expanding network of grade-separated flyovers and elevated structures at major intersections, notably at Koyambedu, Padi, and Madhavaram, generates distinct edge-space conditions at highway junctions. The Outer Ring Road, completed in two phases with Phase II inaugurated in February 2021, spans 62 km along the periphery of the Chennai Metropolitan Area, connecting NH 45, NH 48, CTH Road, and NH 16.

The generation of highway edge spaces across this network results from road widening, flyover construction, grade separation, and the creation of service roads, all of which produce narrow, publicly owned strips of land along corridor margins that are not designated for any active use. These edge spaces vary significantly in their physical characteristics across the GCC highway network. Width ranges from narrow strips of 3–5 m along peri-urban corridor margins to expansive reserves exceeding 80 m adjacent to industrial highway frontages. Physical accessibility conditions range from open verges with direct pedestrian access to completely enclosed strips bounded by compound walls, crash barriers, and drainage channels. Ecological condition, assessed through NDVI in this study, ranges from near-barren surfaces (NDVI < 0.1) in newly widened industrial corridor edges to moderately vegetated strips (NDVI > 0.2) at established flyover junctions with legacy planting. Urban context ranges from dense residential adjacency in inner-city corridors to peri-urban and industrial settings in outer highway zones.

This variation across the four dimensions of width, accessibility, ecological condition, and urban context is not incidental: it reflects the range of morphological conditions that any city-scale NBS suitability framework for highway edge spaces must address. Rather than treating all edge spaces as a single undifferentiated category, this study uses this four-dimensional variation as the basis for a typology of five distinct highway edge space conditions, each associated with a specific NBS intervention category (Section 4.4). Five representative sites Padi (Grand Northern Trunk Road), Villivakkam (Grand Northern Trunk Road), Jafferkhanpet (highway-adjacent wedge space), Koyambedu (flyover junction edge), and Madhavaram (Grand Northern Trunk Road) were selected to cover the full morphological range of this typology within the GCC boundary and subjected to detailed site-scale analysis to validate the NBS Suitability Index scoring derived from GIS data (Section 5.4)

4. Methodology

4.1. Research design and analytical framework

A GIS-based multi-layer spatial analysis framework is developed in this study to identify Priority Intervention Corridors wards within the Greater Chennai Corporation boundary where the reclamation of highway edge spaces through NBS-led urban greening would simultaneously address green space deficit and urban liveliness deficit. The framework is structured around three sequential analytical components: a green space deficit assessment at ward level, a liveliness potential assessment using a composite proxy index, and a spatial overlay integrating both assessments with an NBS priority layer to identify convergence zones. All spatial analysis is conducted in QGIS 3.28 using vector-based methods. GIS-based urban green infrastructure assessments have increasingly incorporated interdisciplinary knowledge systems integrating ecology, urban planning, and sociology to construct evaluations that combine ecosystem functions with residents' needs, providing a more comprehensive picture of urban development priorities.The framework developed here follows this interdisciplinary orientation, combining urban greening deficit measurement with social liveliness indicators to produce a spatially explicit planning output.

The analytical unit throughout is the administrative ward, of which 200 are contained within the GCC boundary. The ward is selected as the unit of analysis because it represents the smallest administratively recognised spatial unit for which population data is available in Chennai, enabling per capita green space calculations to be made at meaningful spatial resolution.

4.2. Data sources

Four primary spatial datasets are used in this study. All datasets are publicly available at no cost and are accessible through open repositories or official municipal portals, consistent with the framework's design principle of replicability in low and middle-income city administrations. Table 2 provides a consolidated summary of all data sources, repositories, access dates, and processing methods applied in QGIS 3.34.

Dataset 1-GCC Ward Boundary and Population Data. Administrative ward boundary shapefiles for the Greater Chennai Corporation (200 wards) were obtained directly from the GCC official municipal portal .Ward-level population figures are drawn from the 2011 Census of

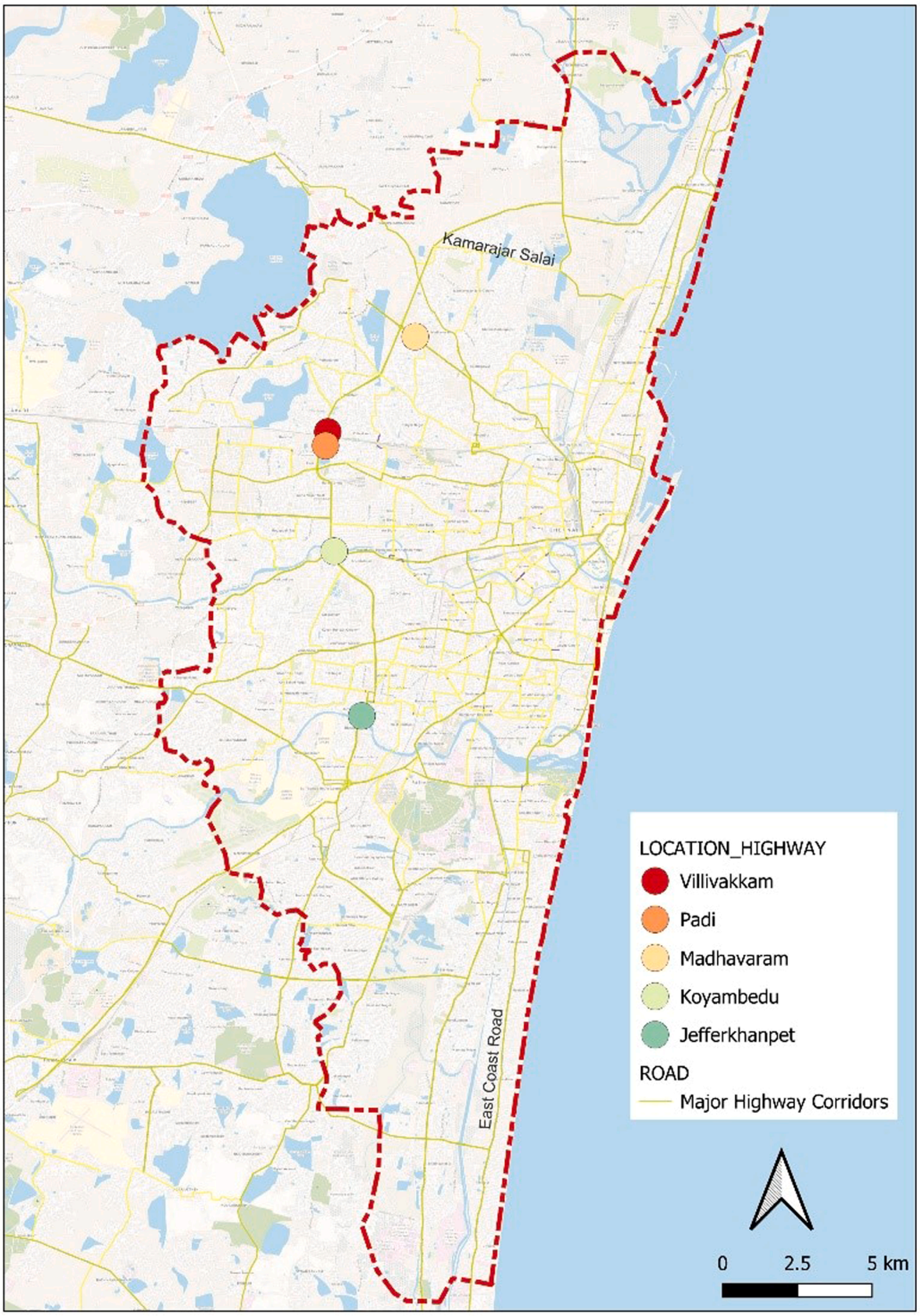

Fig. 1. Major highway corridors within the Greater Chennai Corporation boundary. The map shows the five primary highway alignments (IRR, OMR, GST Road, Poonamallee High Road, and Anna Salai) and the locations of the five representative sites selected for detailed site-scale NBS suitability analysis. Base map: OpenStreetMap contributors; highway corridor delineation:Author.

Table 2
Summary of datasets, sources, repositories, access dates, and processing methods used in.

Dataset	Source	Repository / Access	Date	Processing in QGIS 3.34
Ward boundaries (200 wards)	Greater Chennai Corporation	chennaicorporation.gov.in	2025	Reprojected to UTM Zone 44N (EPSG:32644); area calculation via field calculator
Ward population data	Census of India 2011 (via GCC)	GCC official portal; censusindia.gov.in	2011	Joined to ward boundary shapefile; per capita calculation via field calculator
Urban green space polygons	OpenStreetMap contributors, 2026	openstreetmap.org; QuickOSM plugin v2.2.0	February 2025	Intersected with ward boundaries; area sum per ward via field calculator
Points of Interest (POIs)	OpenStreetMap contributors, 2026	openstreetmap.org; QuickOSM plugin v2.2.0	February 2025	Merged to single POI layer; Count Points in Polygon tool for density per ward
Highway edge space polygons	Manual digitisation; Kailash and Banu Chitra (2025)	Google Earth Pro (Airbus, 2026); QGIS 3.34	February 2025	Polygon import to QGIS; attribute assignment; Buffer tool for highway edge proximity calculation

India, which remains the most recent enumeration available at ward level for Chennai. No population projection to a later year was applied; all per capita calculations in this study are based on 2011 Census ward population figures. This is acknowledged as a limitation in Section 6.4, as population redistribution since 2011 may affect the relative ranking of ward-level deficit scores, though it does not affect the spatial logic of the framework.

Dataset 2 -Urban Green Space Polygons (OpenStreetMap). Existing urban green space polygons within the GCC boundary were extracted from OpenStreetMap (OSM) using the QuickOSM plugin (version 2.2.0) in QGIS 3.34, downloaded in February 2025 (OpenStreetMap contributors, 2026; https://www.openstreetmap.org). Features were extracted under the following OSM key-value pairs: leisure = park, leisure = garden, landuse = grass, and landuse = forest. As OSM is a continuously updated, contributor-maintained database, extracted data reflects contributor mapping status as of the download date and is subject to the coverage limitations discussed in Section 6.4.

Dataset 3 - Points of Interest (OpenStreetMap). Points of Interest (POIs) representing urban activity including shops, health facilities, educational institutions, and public transit stops were downloaded from OpenStreetMap using the same QuickOSM extraction method, downloaded in February 2025.POIs from all relevant OSM categories were merged into a single layer for density calculation. The known limitations of OSM POI coverage in Indian urban contexts, including the systematic underrepresentation of informal commercial activity, are discussed in Section 6.4.

Dataset 4 - Highway Edge Space Polygon Layer. Highway edge space polygons across the GCC boundary were manually digitised against high-resolution satellite imagery in Google Earth Pro (imagery date: February 5, 2025; attribution: Airbus, 2026) at an eye altitude of approximately 624 m, providing sufficient resolution to delineate individual edge space boundaries, measure approximate widths, and distinguish surface conditions. Digitised polygon boundaries were imported into QGIS 3.34 for attribute assignment and spatial analysis. The polygon boundaries used in this study are consistent with the residual space mapping reported in Kailash and Banu Chitra (2025), in which the complete GCC infrastructure-induced residual space inventory of which highway edge spaces form a subset was mapped and validated. The inherent subjectivity of manual digitisation is acknowledged: boundary delineation decisions, particularly at ambiguous edge transitions between highway reserve land and adjacent footpath or service road surfaces, introduce a degree of interpreter variability. This limitation is discussed further in Section 6.4.

4.3. Highway edge space typology mapping

4.3.1. Satellite imagery and digitisation procedure

Highway edge spaces across the GCC boundary were identified and mapped as polygon features through manual digitisation in Google Earth Pro against high-resolution satellite imagery (imagery date: February 5, 2025; attribution: Airbus, 2026) at an eye altitude of approximately 624 m, equivalent to approximately 1:2000 mapping scale. At this scale, individual edge space boundaries, approximate widths, surface conditions, and the presence of physical barriers such as compound walls, crash barriers, and drainage channels are visually distinguishable. Digitised polygon boundaries were subsequently imported into QGIS 3.34 for attribute assignment, area calculation, and integration into the spatial analysis framework. The polygon layer used in this study is consistent with the infrastructure-induced residual space inventory reported in Kailash and Banu Chitra (2025), in which highway edge spaces within the GCC boundary were mapped and validated as a component of a broader residual space typology.

The inherent subjectivity of manual digitisation is acknowledged as a methodological limitation. Boundary delineation decisions particularly at ambiguous transitions between highway reserve land and adjacent footpath, service road, or drainage corridor surfaces involve interpreter judgement that may introduce variability between operators. No formal inter-rater reliability assessment was conducted. This limitation is consistent with manual digitisation practice in comparable urban residual space mapping studies and does not affect the classification or scoring procedures, which are based on measured attributes of the digitised polygons rather than on boundary precision alone. Future applications of the framework would benefit from field-verified boundary demarcation for high-priority sites identified through the NBS-SI scoring

4.3.2. Four-dimension sub-classification framework

Each digitised highway edge space polygon is sub-classified along four independent dimensions, each of which captures a distinct determinant of NBS implementation feasibility. This four-dimension approach is grounded in the site assessment principles established in urban morphology and landscape planning literature: spatial extent and geometry determine the structural capacity for planting (Yang and Li, 2016); physical accessibility determines whether NBS interventions can be maintained and accessed by adjacent populations (Karacor and Akcam, 2016); ecological condition determines the baseline from which NBS must work and the restoration effort required (Forman et al., 2003); and urban context determines the social function and community benefit that NBS should prioritise (Choi, 2017; Kamalipour and Peimani, 2015;Jacobs, 1961).

Dimension 1 -Edge Width. Edge width is measured as the average unobstructed width of each polygon in metres, derived from QGIS field calculator measurements on the imported polygon layer. Three width classes are defined: Narrow (3–5 m), Moderate (5–10 m), and Wide (10 m+). The 3 m minimum threshold reflects the minimum unobstructed width required for single-row tree planting in highway corridor conditions, based on Indian Roads Congress guidelines for roadside tree planting (IRC: 12-2009) and corroborated by comparable urban greening standards applied in Chennai's road development context (CMDA, 2008). Edges narrower than 3 m are not included in the typology as they do not meet the minimum physical threshold for any structural NBS intervention.

Dimension 2 -Physical Accessibility. Physical accessibility is recorded as a three-category attribute -No Access, Limited Access, or Moderate Access - assessed through visual interpretation of satellite

imagery and confirmed through field verification for the five representative sites (Section 5.4). No Access indicates the edge space is fully enclosed by compound walls, crash barriers, drainage channels, or other continuous physical barriers with no pedestrian entry point visible. Limited Access indicates partial enclosure or intermittent barriers with at least one identifiable entry point. Moderate Access indicates open edges with direct pedestrian connectivity from adjacent footpaths or streets. Field verification across the five study sites revealed that GIS-derived accessibility scores systematically overestimated access conditions due to the inability of satellite imagery to detect low-height barriers, temporary obstructions, and informal enclosures, a finding discussed in Section 5.4 and 6.4.

Dimension 3 - Ecological Condition (NDVI). Ecological condition is assessed using the Normalised Difference Vegetation Index (NDVI), extracted from Sentinel-2 satellite imagery at 10 m resolution for each polygon using the Zonal Statistics tool in QGIS 3.34. NDVI median values per polygon are classified into three ecological condition categories: Low Green (NDVI < 0.15), Medium Green (NDVI 0.15–0.20), and High Green (NDVI > 0.20). These thresholds are consistent with NDVI classification ranges established for urban vegetation assessment in South Asian tropical contexts, where bare or sealed surfaces typically return NDVI values below 0.1, sparse herbaceous cover falls between 0.1 and 0.2, and established tree and shrub cover exceeds 0.2 (Sathyakumar et al., 2020; Lahoti et al., 2019). NDVI-based ecological scoring provides an objective, remotely-assessable measure of baseline vegetation condition that does not require field survey, maintaining the framework's data accessibility principle.

Dimension 4 - Urban Context. Urban context is recorded as a categorical attribute describing the dominant land use of the area immediately adjacent to each highway edge space: Residential, Industrial, Mixed (residential and commercial), or Peri-urban. This attribute is assigned through visual interpretation of satellite imagery cross-referenced with OSM land use data. Urban context determines the primary social function that NBS interventions should target -community amenity and active recreation in residential contexts, ecological buffering and phytoremediation in industrial contexts, environmental quality improvement in mixed contexts, and ecological seeding and long-term succession in peri-urban contexts (Kabisch et al., 2017; Raymond et al., 2017).

4.3.3. *Five highway edge space typologies and NBS intervention categories*

The combination of the four sub-classification dimensions produces five distinct highway edge space typologies, each associated with a specific NBS intervention category and a recommended NBS package (Table 3). The five typologies cover the full morphological range of highway edge conditions encountered across the GCC boundary and are designed to be directly actionable: each typology maps to a specific NBS intervention category that reflects the physical constraints, ecological baseline, and social context of that condition. The typologies are not mutually exclusive in principle but represent the dominant condition combinations observed across the mapped polygon inventory.

4.3.4. *NBS suitability index (NBS-SI) scoring*

Each highway edge space polygon is assigned a composite NBS Suitability Index (NBS-SI) score derived from four sub-classification dimensions. Each dimension contributes a sub-score on a 1–3 scale, and the four sub-scores are summed to produce a total NBS-SI score ranging from 4 (lowest suitability) to 12 (highest theoretical suitability). In practice, the five typologies identified across the GCC boundary yield NBS-SI scores from 3 to 9, reflecting the constraint that the lowest-scoring dimension (typically accessibility or ecological condition) caps the overall suitability for any given site. The scoring logic is as follows: Width - Narrow (1), Moderate (2), Wide (3); Accessibility - No Access (1), Limited Access (2), Moderate Access (3); Ecological Condition - Low Green (1), Medium Green (2), High Green (3); Urban Context - Peri-urban (1), Industrial (2), Mixed or Residential (3).

NBS-SI scores categorise the 82 Priority Intervention Corridors into two operational tiers. Tier 1 corridors, with scores of 7–9, are sites ready for immediate NBS deployment using standard planning and engineering methods. Tier 2 corridors, scoring 5–6, require preparatory actions such as improving access or establishing ecological baselines before full implementation. Sites with scores of 3–4, mainly of the T5 Passive NBS type, are targeted for long-term ecological seeding and monitoring instead of immediate intervention.

4.3.5. *NBS decision matrix and framework transferability*

The NBS-SI scoring system is designed as a dynamic decision matrix rather than a fixed threshold tool. While the width thresholds applied in this study draw on Indian Roads Congress guidelines (IRC: 12-2009) and Chennai-specific road development standards (CMDA, 2008), the four-dimension scoring logic and the five typology categories are not anchored to Indian national standards alone. Planners applying the framework in other urban contexts - including metropolitan areas in Latin America, Southeast Asia, or sub-Saharan Africa -can recalibrate the width thresholds to local road reserve standards, adjust the NDVI ecological condition thresholds to local climate and vegetation density norms, and reweight the urban context categories to reflect local planning priorities, without altering the underlying four-dimension classification structure.

The only fixed requirement for application in a new city context is the availability of: (1) a digitised highway edge space polygon layer with measurable width attributes; (2) satellite imagery for NDVI extraction; (3) field or imagery-based accessibility assessment; and (4) land use data for urban context classification. All four inputs are obtainable from open data sources in most Global South urban contexts, maintaining the framework's core principle of accessibility to planning departments with limited technical and financial capacity.

4.4. *Green space deficit assessment*

Existing urban green space coverage across the GCC boundary is assessed at ward level by intersecting the OpenStreetMap green space polygon layer with the ward boundary shapefile. The total green space

Table 3
Five highway edge space typologies, sub-classification conditions, NBS Suitability Index scores, NBS intervention categories, and recommended NBS intervention.

Typology	Condition Combination	NBS-SI Score	Priority	NBS Function	Recommended NBS Package	Representative Site
T1	Wide (10m+) + No Access + Low Green + Industrial	5	Medium	Remediation NBS	Phytoremediation planting + soil restoration + dense tree belt for air quality	Site 1: Padi, GNT Road
T2	Wide (10m+) + Moderate Access + Low Green + Dense Residential	7	High	Active NBS	Multi-layer tree belt + bioswale + shaded walking edge + community greening	Site 2: Villivakkam, GNT Road
T3	Moderate (5–10m) + Limited Access + Medium Green + Res/Mixed	5	Medium	Enhancement NBS	Infill native planting + stormwater swale + noise buffer vegetation	Site 3: Jafferkhanpet wedge space
T4	Wide (10m+) + Limited Access + High Green + Mixed	7	High	Conservation NBS	Ecological conservation + biodiversity enrichment + pollinator pathway	Site 4: Koyambedu junction/flyover edge
T5	Narrow (3–5m) + No Access + Low Green + Peri-urban	3	Low	Passive NBS	Single-row tree planting + grass strip for erosion control	Site 5: Madhavaram, GNT Road

area within each ward is calculated using the field calculator in QGIS. Per capita green space is then derived by dividing total ward green space area in square metres by ward population, producing a gs_per_cap value for each ward. Wards are classified into three deficit categories high, medium, and low deficit using Natural Breaks (Jenks) classification. The Government of India recommended standard of 10–12 sq. m. of green space per capita is applied as the reference threshold against which ward-level values are assessed (Govindarajulu, 2014). A deficit score (deficit_sc) is subsequently derived by inverting the normalised per capita green space value so that wards with the least green space receive the highest deficit score, in preparation for the spatial overlay.

4.5. Liveliness potential assessment

Urban liveliness potential is assessed at ward level using a composite index constructed from three proxy variables - POI density, population density, and highway edge proximity - each of which is independently calculated and normalised to a 0–1 scale before combination. The index does not measure street-level urban activity directly; it approximates the spatial conditions under which urban liveliness is most concentrated and most deficient, enabling wards where NBS-led greening of highway edges would produce the greatest liveliness co-benefit to be identified at city scale.

POI density is calculated as the number of merged OpenStreetMap POIs per square kilometre per ward using the Count Points in Polygon tool in QGIS 3.34. POI density is used as a proxy for functional diversity - the concentration of shops, health facilities, educational institutions, and transit stops that generate the variety of uses Kamalipour and Peimani (2015) identifies as the primary driver of street-level activity. Population density is calculated as ward population per square kilometre using ward area derived from the reprojected shapefile. Population density is included as a proxy for the concentration of residents whose daily liveliness conditions are most directly affected by the absence of accessible urban greening -densely populated wards contain the greatest number of residents who stand to benefit from NBS-led improvement to the pedestrian environment (Kabisch et al., 2017). Highway edge proximity is measured as the percentage of each ward's area that falls within a 400-metre buffer of the highway edge typology layer, using the standard pedestrian catchment distance established in urban green space accessibility research (Kaźmierczak, 2013). This variable measures the spatial coverage of highway edge influence within each ward - the proportion of ward area where residents are within walking distance of a highway edge space and therefore directly exposed to both the environmental burden and the potential greening benefit of that edge. The term highway edge proximity is used in preference to walkability throughout this study, as the variable does not measure pedestrian-friendliness, path quality, or connectivity - attributes associated with established walkability metrics - but rather the spatial reach of highway edge spaces within the ward boundary.

The three variables are assigned equal weight in the composite index. This decision is justified on three grounds. First, each variable captures a theoretically distinct and non-redundant dimension of liveliness deficit: functional diversity (POI density), residential concentration (population density), and spatial exposure to highway edge conditions (highway edge proximity). No single variable subsumes the information content of the others, and no empirical basis exists within the Chennai dataset for asserting that one dimension is systematically more predictive of liveliness deficit than the others across all 188 wards included in the analysis. Second, equal weighting is consistent with composite index construction practice in urban vitality and green space accessibility research where variable importance is theoretically contested and sensitivity testing is used to assess robustness (Texier et al., 2018; Aghaloo et al., 2024; Cohen-Shacham et al., 2016). Third, equal weighting maximises the transparency and replicability of the index for planning practitioners in other city contexts, where locally specific variable weightings would undermine the framework's transferability without a city-specific empirical calibration study. The sensitivity of the composite index to alternative weighting schemes - including population-dominant and POI-dominant configurations - is acknowledged as a direction for future methodological refinement.

Each variable is normalised using minimum-maximum normalisation:

$$Normalisedvalue = (x - minimum)/(maximum - minimum)$$

The three normalised variables are averaged to produce a final Liveliness Potential Score per ward:

$$LivelinessScore = (POIdensitynorm + Populationdensitynorm + HighwayEdgeProximitynorm)/3$$

Wards are classified into three liveliness potential categories - high, medium, and low - using Natural Breaks (Jenks) classification. The Jenks method is selected because the distribution of liveliness potential scores across the 188 included wards is non-normal and multimodal, with natural clustering patterns that are better captured by a data-driven classification than by equal-interval or quantile methods, which would impose artificial boundaries on the observed distribution. It is acknowledged, however, that Jenks classification produces thresholds that are specific to the data distribution of the sample and are therefore not directly transferable to other cities as fixed benchmarks. Planners applying the framework in other urban contexts should generate Jenks thresholds from their own ward-level score distributions, or substitute an alternative classification method - such as standard deviation or k-means clustering - if local data distributions differ substantially from the Chennai pattern. This limitation does not affect the internal validity of the priority identification procedure within the GCC study area, where the Jenks thresholds correctly reflect the natural structure of the score distribution.

Wards falling in the low liveliness potential category are identified as the population most in need of urban greening interventions that generate the spatial conditions associated with active public life, pedestrian movement, and social engagement (Montgomery, 1998; Jacobs, 1961). Twelve wards record zero POI counts under the OpenStreetMap extraction and are excluded from the composite score calculation, as insufficient POI data precludes a reliable liveliness estimate. These excluded wards are displayed as a separate not-significant category in the results maps and their exclusion is discussed as a data limitation in Section 6.4.

4.6. NBS priority mapping

NBS priority points across the GCC highway edge network are drawn from the residual space mapping reported in Kailash and Banu Chitra (2025), in which infrastructure-induced residual spaces across the GCC boundary -including highway edge spaces, metro viaduct understories, and junction wedge spaces - were mapped, classified, and assessed for NBS implementation suitability based on corridor conditions and residential adjacency. The highway edge space subset of that

inventory, comprising the polygon layer used in this study, includes spatial records of locations assessed as suitable for NBS implementation based on measurable site-level criteria including edge width, surface condition, physical accessibility, and proximity to residential populations. This layer serves in the present study as the third input in the spatial overlay, grounding the convergence analysis in site-level NBS suitability assessments derived from field-informed mapping rather than ward-level estimation alone.

4.7. Spatial overlay and priority intervention corridor identification

The three analytical outputs green space deficit score, liveliness potential score, and NBS priority points are integrated through a two-stage spatial overlay to identify Priority Intervention Corridors. In the first stage, a combined priority score is calculated for each ward:

$$PriorityScore = (DeficitScore + (1 - LivelinessScore))/2$$

This formulation combines high green space deficit with low liveliness potential, producing the highest priority scores in wards where urban greening need and urban liveliness deficit are simultaneously concentrated. Wards are classified into three priority categories using Natural Breaks (Jenks) classification. In the second stage, high-priority wards are intersected with the NBS priority point layer using the Select by Location tool in QGIS, retaining only those high-priority wards that contain at least one NBS priority point. The resulting selection exported as convergence_zones.shp constitutes the Priority Intervention Corridors: wards where urban greening through highway edge NBS reclamation is both spatially warranted by deficit and liveliness conditions and grounded in site-level NBS suitability. Using the GIS platform, a map of spatial suitability for urban green space is developed by weighted overlay analysis, providing decision-makers with a precise reference for selecting suitable sites. The framework developed here adapts this approach to the specific spatial logic of highway edge corridor reclamation, producing a result that is both analytically rigorous and directly actionable for urban greening policy.

5. Results

5.1. Overview

Three spatial layers are produced through the analytical framework: a ward-level green space deficit map, a ward-level liveliness potential map, and a Priority Intervention Corridor map identifying wards where both conditions converge with NBS-suitable highway edge locations. The results are presented in sequence below, with each layer described in terms of its spatial distribution across the GCC boundary before the overlay findings are reported. All three maps are presented as Figs. 1–3.

5.2. Green space deficit across GCC wards

The ward-level assessment of per capita green space reveals an acute and near-universal green space deficit across the Greater Chennai Corporation boundary (Fig. 2). Of the 200 wards assessed, the overwhelming majority are classified in the high deficit category under the Natural Breaks (Jenks) classification, with per capita green space values falling substantially below the Government of India recommended standard of 10–12 sq. m. per capita (Govindarajulu, 2014). Medium deficit wards are scattered across the mid-city zone with no discernible spatial clustering. Only a small number of wards record low deficit values, and these are concentrated in two spatial locations: a cluster in the western mid-city corresponding to the Guindy National Park and its adjacent institutional green reserve, and a secondary cluster in the south where road-side vegetation and open land associated with the GST Road corridor contribute marginally higher green coverage.The spatial pattern confirms that green space provision in Chennai is not merely inadequate at the city scale it is structurally unequal, with the burden of deficit falling most heavily on high-density residential wards across the northern, eastern, and central zones. The Guindy exception is notable precisely because it underscores the rule: institutional green reserves of this scale are rare within the GCC boundary, and their contribution to the overall city average conceals the severity of deficit in the remaining wards. No ward outside the Guindy cluster approaches the GoI recommended standard.This finding substantiates the identification of highway edge spaces as a priority opportunity for urban greening. Given the absence of available land for conventional park creation within the built fabric of dense Chennai wards, the reclamation of existing edge spaces through NBS-led interventions street tree planting, linear parkways, vegetated medians, and roadside shrub corridors represents the most spatially viable mechanism for incrementally expanding per capita green space at ward level (Govindarajulu, 2014; Lahoti et al., 2019; Lehmann, 2021).

5.3. Liveliness potential across GCC wards

The composite liveliness potential index, constructed from normalised POI density, population density, and walkability to highway edge spaces, produces a spatially differentiated pattern across the GCC boundary (Fig. 3). High liveliness potential is concentrated in the central core of the city wards corresponding to Anna Nagar, Nungambakkam, Teynampet, and Mylapore where POI density is highest, population is concentrated, and proximity to the highway corridor network is greatest. These wards represent locations where the conditions associated with active public life, pedestrian movement, and social engagement are already present or latent (Karacor and Akcam, 2016; Mastervich et al., 2024; McConnell and Walls, 2005). Low liveliness potential is concentrated in two distinct zones. The first is the northern corridor wards in Tiruvottiyur, Manali, and Madhavaram where industrial land use, lower POI density, and distance from major activity corridors combine to produce consistently low scores. The second is the eastern coastal corridor along the OMR, where recent residential development has outpaced the provision of walkable street-level activity, producing wards with moderate-to-high population density but low amenity and transit access. A number of peripheral western wards also record low liveliness scores, reflecting incomplete urban fabric and limited street-level activation.Twelve wards record zero POI counts under the OpenStreetMap data extraction. These wards are assigned grey classification in Fig. 1, and excluded from the composite score, as POI data is insufficient to produce a reliable liveliness estimate. This is acknowledged as a data limitation and discussed further in Section 6. The spatial distribution of liveliness potential is consistent with the theoretical relationship between green space access, tree canopy cover, and the quality of public space identified in the literature. Wards with low liveliness potential are precisely those where the absence of walkable, vegetated public space is most acutely felt and where highway edge reclamation through NBS could most meaningfully contribute to the conditions for street-level engagement and pedestrian comfort (Iungman et al., 2023).

5.4. Site-scale NBS suitability verification: five representative highway edge sites

The ward-level spatial overlay identifies where NBS intervention is strategically warranted across the GCC boundary. To evaluate whether the NBS-SI scoring framework produces reliable and implementable outputs at the site scale, five representative highway edge spaces were selected - one per typology -and subjected to a six-step detailed site analysis combining GIS measurement, NDVI extraction, and field verification. The five sites - Padi (T1), Villivakkam (T2), Jafferkhanpet (T3), Koyambedu (T4), and Madhavaram (T5) - were selected to cover the full morphological range of conditions encountered across the 82 Priority Intervention Corridors, including linear verge edges on a national highway (Sites 1, 2, 5), a highway-adjacent wedge space at a road

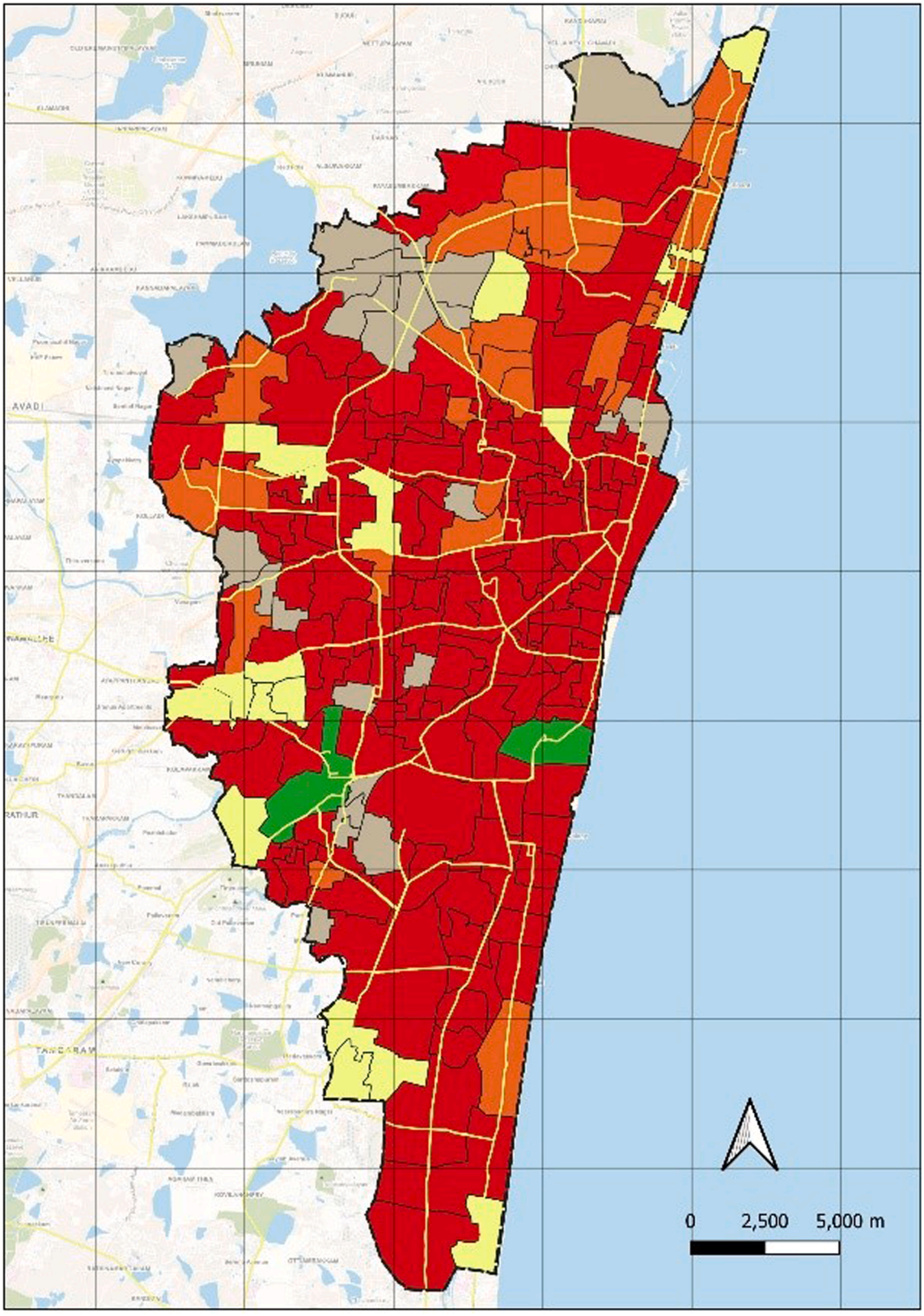

Fig. 2. The map shows that the majority of wards across the Greater Chennai Corporation boundary fall under high green space deficit (red), with medium deficit wards (yellow-green) scattered across the mid-city zone and low deficit (dark green) restricted to two small clusters in the western mid-city corresponding to the Guindy National Park reserve.

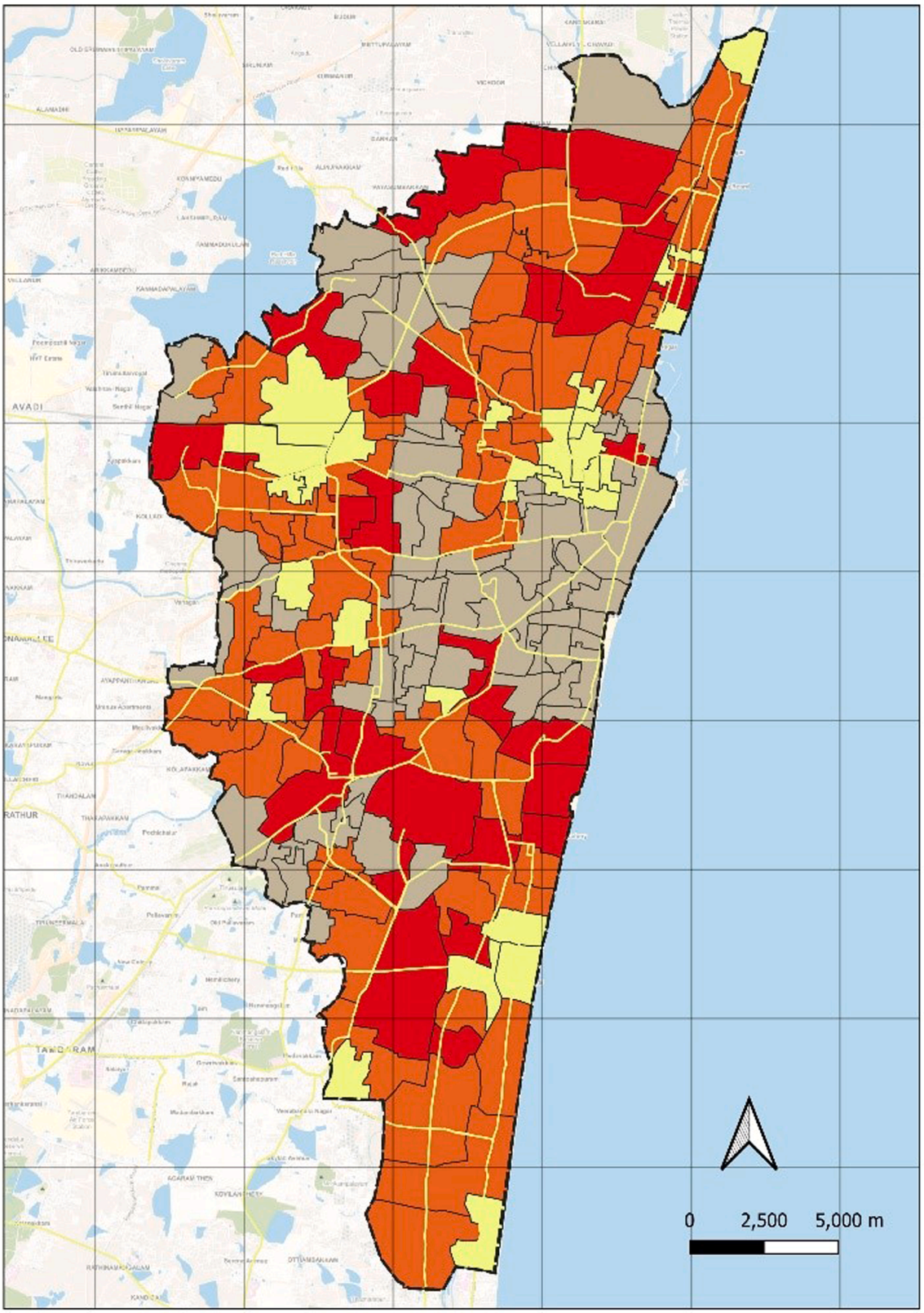

Fig. 3. The map shows high liveliness potential (red) concentrated in the northern and central wards, medium liveliness (orange) spread across the mid-city, and low liveliness (yellow) and not-significant wards (grey-brown) distributed across the peripheral northern, eastern coastal, and southern zones along the highway edge corridors (yellow lines).

Table 4
Site characteristics, GIS-derived NBS-SI scores, field-verified scores, and recommended NBS packages for five representative highway edge sites.

Parameter	Site 1 Padi (T1)	Site 2 Villivakkam (T2)	Site 3 Jafferkhanpet (T3)	Site 4 Koyambedu (T4)	Site 5 Madhavaram (T5)
Typology	T1 -Remediation NBS	T2 - Active NBS	T3 -Enhancement NBS	T4 -Conservation NBS	T5 - Passive NBS
Width (m)	86 m (Wide)	11–18 m (Wide)	7 m avg (Moderate)	40–70 m (Wide)	3–5 m (Narrow)
Area (m^2)	11,857	3,525	757	6,886	417
NDVI Median	0.183 (Low Green)	0.083 (Low Green)	0.124 (Low-Medium)	0.219 (High Green)	0.143 (Low Green)
Elevation (m)	12–13	10–14	7–10	7–9	10 (flat)
Population within 300 m	69,532	41,640	1,06,208	63,059	31,083
Population within 500 m	1,54,799	1,10,540	2,74,660	1,80,291	77,330
Urban Context	Industrial	Dense Residential	Residential + Mixed	Mixed	Peri-urban
GIS-derived NBS-SI Score	**5**	**7**	**5**	**7**	**3**
Field-verified NBS-SI Score	**5**	**6**	**4**	**6**	**3**
NBS Function Confirmed	Yes	Yes (minor adjustment)	Yes (minor adjustment)	Yes (minor adjustment)	Yes
Access Score GIS-Field Match	Match	GIS overestimate	GIS overestimate	GIS overestimate	Match

junction (Site 3), and a flyover junction edge space (Site 4). All five sites are located within wards identified as Priority Intervention Corridors in the ward-level overlay.

The six-step site analysis procedure applied to each site is: (1) polygon boundary delineation and width measurement in QGIS 3.34; (2) NDVI extraction from Sentinel-2 imagery using the Zonal Statistics tool; (3) physical accessibility assessment from satellite imagery and field observation; (4) urban context classification from satellite imagery cross-referenced with OSM land use; (5) NBS-SI score calculation from GIS-derived inputs; and (6) field verification of GIS-derived scores against on-site physical conditions. Steps 1–5 constitute the GIS-derived score; Step 6 constitutes the field-verified score. Discrepancies between the two scores are recorded and analysed to identify systematic limitations of the GIS-based approach Table 4..

5.4.1. Site-by-site findings

Site 1 -Padi, Grand Northern Trunk Road (T1: Remediation NBS). The Padi site presents the widest edge space in the study at 86 m, covering 11,857 m^2 in an industrial corridor context. Despite its substantial spatial extent, the site records an NDVI median of 0.183- classifying as Low Green - because field observation confirmed that the apparent vegetation cover consists predominantly of invasive and degraded species rather than ecologically functional planting. This finding illustrates a critical limitation of NDVI-based ecological scoring: visually green surfaces can generate moderate NDVI values while providing minimal ecological function. The GIS-derived NBS-SI score of 5 and field-verified score of 5 are in agreement; the No Access condition was confirmed by continuous compound walling along the site perimeter. The recommended NBS package - phytoremediation planting, soil restoration, and a dense tree belt for air quality - is appropriate given the industrial context, with 69,532 residents within 300 m and 1,54,799 within 500 m bearing direct exposure to industrial corridor emissions.

Site 2 -Villivakkam, Grand Northern Trunk Road (T2: Active NBS). The Villivakkam site is a 3,525 m^2 wide edge space (11–18 m) in a dense residential context, with the highest immediate NBS intervention potential of all five sites. The GIS-derived score of 7 reflects Wide width, Low Green NDVI (0.083), and a GIS-assessed Moderate Access condition based on the absence of continuous barriers in the satellite imagery. Field verification, however, identified intermittent railings and informal enclosures along the site frontage that reduce effective accessibility below the GIS-assumed level, lowering the field-verified score to 6. Despite this access score correction, the NBS function- Active NBS comprising a multi-layer tree belt, bioswale, shaded walking edge, and community greening - is confirmed as appropriate. With 41,640 residents within 300 m and 1,10,540 within 500 m in a dense residential setting, this site represents the highest community benefit potential per unit area of any site in the study.

Site 3 -Jafferkhanpet Wedge Space (T3: Enhancement NBS). The Jafferkhanpet site is the smallest in the study at 757 m^2, representing a highway-adjacent wedge space with moderate average width of 7 m in a residential and mixed-use context. The NDVI median of 0.124 places the site at the low-to-medium boundary of ecological condition. GIS imagery indicated Limited Access, but field verification found additional physical barriers - concrete retaining elements and informal enclosures -that restrict access more severely than the satellite assessment suggested, reducing the field-verified score from 5 to 4. The recommended Enhancement NBS package - infill native planting, a stormwater swale, and noise buffer vegetation - is confirmed as appropriate given the site's moderate width and its location within a ward recording 1,06,208 residents within 300 m, the highest residential exposure of any site in the study.

Site 4 - Koyambedu Junction/Flyover Edge (T4: Conservation NBS). The Koyambedu site is the ecologically richest of the five, recording the highest NDVI median at 0.219 - classifying as High Green - across a wide edge space of 40–70 m covering 6,886 m^2 at a major highway junction and flyover. It is important to note that the Koyambedu NDVI value is subject to a context-specific limitation: the Cooum River runs adjacent to the site, and water body pixels within the Sentinel-2 extraction zone dilute the NDVI median, meaning the actual vegetation cover of the edge space may be higher than the extracted value indicates. Field verification confirmed the presence of established tree cover and diverse ground-level vegetation consistent with a High Green classification. The GIS-derived score of 7 is corrected to 6 in field verification due to the physical barriers -crash barriers and traffic infrastructure - that restrict pedestrian access to the interior of the edge space despite its apparent openness in satellite imagery. The Conservation NBS package- ecological conservation, biodiversity enrichment, and a pollinator pathway - is confirmed as appropriate and represents the most ecologically significant intervention opportunity in the study.

Site 5 - Madhavaram, Grand Northern Trunk Road (T5: Passive NBS). The Madhavaram site is the smallest and lowest-scoring in the study, with a narrow width of 3–5 m covering only 417 m^2 in a peri-urban context. The NDVI median of 0.143 confirms a Low Green ecological condition. Both the No Access condition and the narrow width were independently confirmed in the field, yielding full GIS-field agreement at a score of 3. No active NBS intervention is considered feasible given the site's current physical constraints. The recommended Passive NBS package - ecological seeding and natural succession monitoring, with the option of a single-row tree planting along the outer edge if a minimum 3 m clearance can be established - is appropriate as a long-term ecological buffering strategy. With only 31,083 residents within 300 m and a peri-urban context, this site represents the lowest immediate community benefit potential, consistent with its Low priority classification.

5.4.2. Comparative analysis: framework performance across five sites

Three findings emerge from the comparative GIS-field analysis across the five sites. First, the NBS function assigned by the framework was confirmed as appropriate in all five sites -three with full agreement and two with minor site-specific adjustments to intervention sequencing. This confirms that the four-dimension typology correctly differentiates highway edge spaces and assigns NBS categories that

reflect ground conditions. Second, the width-based typology score achieved 100% GIS-field match across all five sites: GIS-measured widths corresponded closely to field measurements in every case, validating width as a reliable, remotely-assessable classification parameter that does not require field survey to score accurately. Third, the accessibility score is the framework's weakest parameter: only 1 of 5 sites showed full GIS-field agreement on the access sub-score. Physical barriers - concrete compound walls, crash barriers, informal railings, and traffic infrastructure - consistently block access that GIS buffer analysis assumes to exist based on spatial proximity alone. This systematic GIS overestimation of accessibility is the primary refinement need identified by the site-scale verification and is discussed further in Section 6.4.

The NDVI-based ecological score performed reliably in four of five sites. The two context-specific limitations identified - NDVI underestimation due to adjacent water body pixels at Koyambedu, and NDVI overestimation of ecological function where vegetation is invasive or degraded at Padi - indicate that NDVI extraction should be complemented by visual imagery interpretation or brief field checking at high-priority sites before final NBS category assignment. These limitations do not invalidate NDVI as a screening parameter at city scale, where its remote-assessability and objectivity outweigh its site-specific imprecision for initial corridor prioritisation purposes.

5.5. Priority intervention corridors

The spatial overlay of green space deficit score, inverted liveliness potential score, and NBS priority points identifies 82 wards as Priority Intervention Corridors across the GCC boundary (Fig. 4). These 82 wards - representing 41% of the 200 GCC wards included in analysis - are locations where high green space deficit and low liveliness potential co-occur simultaneously, and where at least one NBS-suitable highway edge location has been identified within the ward boundary. Of the 200 total GCC wards, 188 were included in the composite scoring; 12 wards with zero OSM POI coverage were excluded from the liveliness index calculation and do not appear in the priority classification.

The 82 Priority Intervention Corridors are not distributed uniformly across the GCC. Four spatial clusters are observable. The largest cluster is concentrated in the northern zone wards along the Grand Northern Trunk Road corridor where industrial land use, high population density, and acute green space shortfall converge. A second cluster follows the eastern coastal corridor along the OMR (Rajiv Gandhi Salai), where rapid residential expansion has produced high population density without commensurate green space or amenity provision. A third cluster is identifiable in the western mid-city zone along the Poonamallee High Road and IRR alignments, where dense residential wards record both low per capita green space and limited walkable public activity space. A fourth, smaller cluster is present in the southern wards along the GST Road alignment. Notably, the central core wards those recording the highest liveliness potential scores are largely absent from the Priority Intervention Corridor map, consistent with the framework's logic that high liveliness wards, even where green space deficit exists, do not meet the dual-condition threshold that triggers priority classification.

To enhance the operational utility of the 82 Priority Intervention Corridors for planning departments, the corridors are further stratified into two implementation tiers based on their NBS-SI scores derived from the highway edge space typology assessment (Section 4.3.4). This stratification addresses the practical concern that identifying 41% of all GCC wards as equally prioritised would scatter decision-maker attention and undermine the framework's usefulness for resource allocation. Tier 1 corridors (NBS-SI score 7–9) represent wards containing highway edge spaces with high immediate NBS intervention potential sufficient width, adequate or correctible access conditions, and appropriate ecological baseline where NBS deployment can proceed with standard planning and engineering processes. Tier 2 corridors (NBS-SI score 5–6) represent wards where highway edge spaces require preparatory measures primarily physical access improvement or ecological baseline establishment before full NBS implementation is feasible Table 5.

The 82 Priority Intervention Corridors collectively span the major highway corridors mapped across the GCC boundary the Grand Northern Trunk Road, IRR, OMR, GST Road, and Poonamallee High Road confirming that the spatial opportunity for NBS-led highway edge reclamation is distributed across the city's radial road network rather than concentrated on any single corridor. This distribution strengthens the case for a corridor-based urban greening strategy in which street tree planting, linear parkways, vegetated buffer zones, and bioswales are deployed as a connected network of green interventions along highway edges, producing cumulative ecological benefit across the city's green infrastructure system (Grabowski et al., 2023; Malaya et al., 2024). The Tier 1 / Tier 2 stratification enables planning departments to sequence NBS investment systematically beginning with Tier 1 corridors where implementation barriers are lowest and community benefit potential is highest, and advancing to Tier 2 corridors as access and ecological preparatory works are completed.

6. Discussion

6.1. The spatial logic of highway edge reclamation

The results of the spatial analysis confirm that highway edge spaces within the Greater Chennai Corporation boundary are a largely underused resource for urban greening, especially as traditional park development is no longer spatially feasible within the city's built environment. The widespread green space deficiency observed across GCC wards, with per capita green space well below the Government of India's recommended standard of 10–12 sq. m. per person (Govindarajulu, 2014; Masiero et al., 2022; Li et al., 2025), cannot be effectively addressed by simply identifying new land parcels in the already dense urban setting. Therefore, reclaiming existing highway edge spaces through Nature-Based Solutions becomes not just an aspirational planning goal but a necessary spatial strategy.

This argument aligns with a growing body of urban forestry literature that considers linear green infrastructure- such as street tree corridors, vegetated road margins, roadside shrub plantings, and highway buffer greening- as the main way to gradually increase green space per person in cities with limited available land (Grabowski et al., 2023; Lahoti et al., 2019; Govindarajulu, 2014). Highway edges are especially suitable for this purpose because they are already publicly owned, run continuously throughout the city, and are accessible to large residential populations- conditions that cannot be matched by creating new parks in dense urban areas (Sanches and Pellegrino, 2016). The 82 Priority Intervention Corridors identified in this study are precisely those wards where this spatial approach is most necessary- where the deficit is largest, liveliness conditions are weakest, and confirmed edge locations suitable for Nature-Based Solutions (nbs) are available.

6.2. Greenness, liveliness, and the role of vegetation in public space

A central argument of this paper is that the reclamation of highway edge spaces through NBS is not only an ecological intervention but also a liveliness intervention that the introduction of tree canopy, ground-level vegetation, and green buffer zones along highway corridors can directly improve the conditions for pedestrian activity, social engagement, and street-level public life in adjacent wards. The spatial correspondence observed between low liveliness potential and high green space deficit across the Priority Intervention Corridors provides spatial evidence in support of this relationship.

This relationship is well established in the urban forestry literature. Tree canopy cover has been shown to reduce thermal discomfort along pedestrian routes, extending the hours and seasons during which street-level activity is viable in hot-climate cities (Iungman et al., 2023). The presence of street trees and roadside vegetation has been associated with increased pedestrian dwell time, reduced traffic noise perception, and

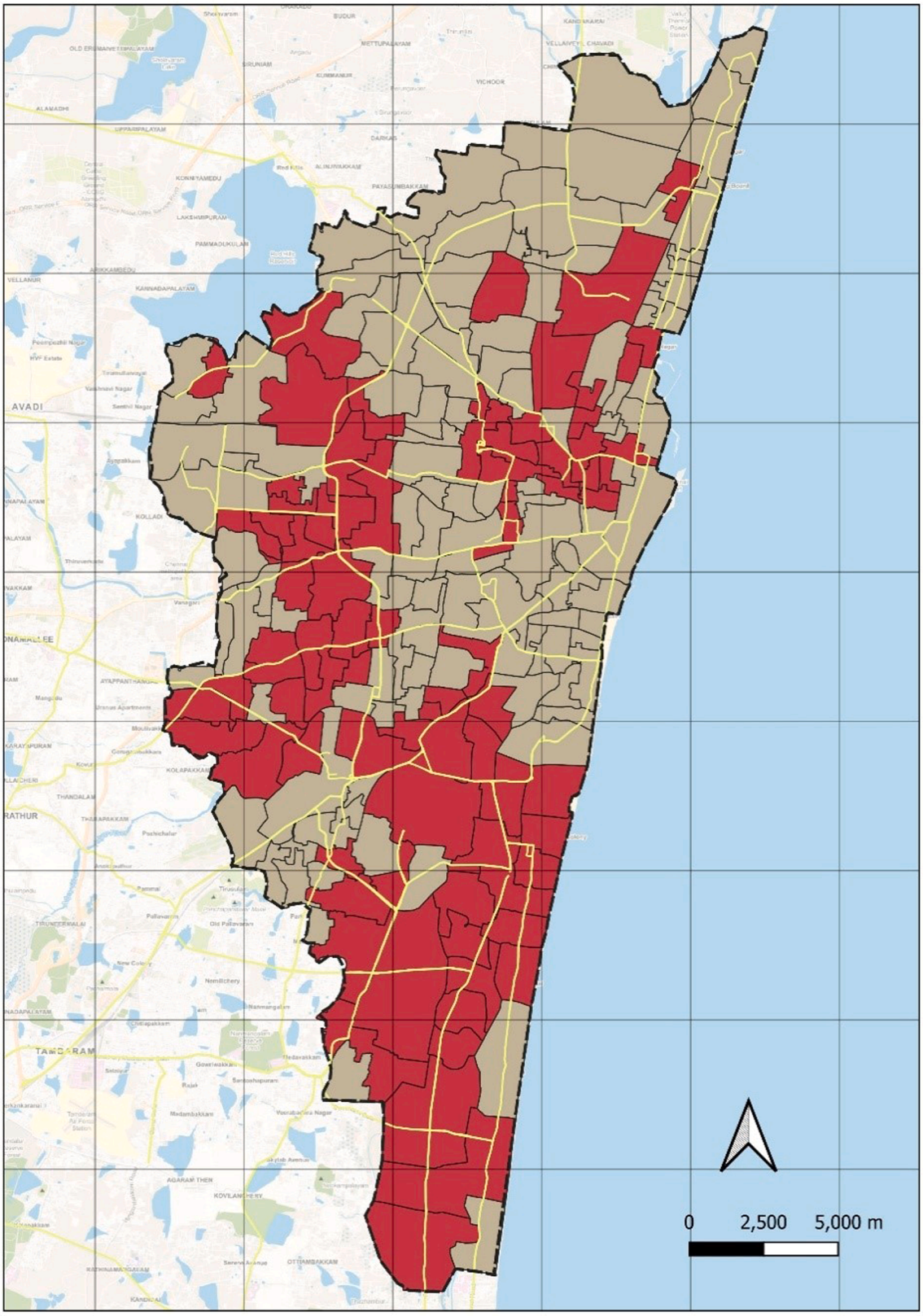

Fig. 4. The map shows that the majority of wards across the Greater Chennai Corporation boundary are identified as High Priority Corridors (red), concentrated particularly across the southern half and mid-city zones, with not-significant wards (grey-brown) distributed primarily across the northern periphery. Yellow lines indicate the highway edge corridor network threading through the priority zones.

Table 5
Tier 1 and Tier 2 stratification of the 82 Priority Intervention Corridors by NBS-SI score range, typology distribution, and implementation action.

Tier	NBS-SI Score Range	Dominant Typologies	Implementation Action	NBS Package
Tier 1 (Immediate Intervention)	**7–9**	T2 (Active NBS) - Wide + Moderate Access + Residential T4 (Conservation NBS) -Wide + Limited Access + High Green	Proceed directly to site-level design and engineering. Field verification of access conditions recommended before implementation. No preparatory measures required.	Multi-layer tree belt + bioswale + shaded walking edge + community greening (T2); Ecological conservation + biodiversity enrichment + pollinator pathway (T4)
Tier 2 (Preparatory Measures Required)	**5–6**	T1 (Remediation NBS) - Wide + No Access + Industrial T3 (Enhancement NBS) - Moderate + Limited Access + Medium Green	Address physical access barriers first. Ecological baseline establishment or soil remediation may be required. Full NBS implementation after preparatory works.	Phytoremediation + soil restoration + dense tree belt (T1); Infill native planting + stormwater swale + noise buffer vegetation (T3)
Long-term (Ecological Seeding Only)	**3–4**	T5 (Passive NBS) - Narrow + No Access + Low Green + Peri-urban	No immediate structured intervention. Monitor for natural succession. Single-row tree planting if 3 m clearance can be established in future.	Ecological seeding + natural succession monitoring

greater willingness to use public space. In the Chennai context where ambient temperatures regularly exceed 35°C and where the absence of shade along highway corridors constitutes a direct barrier to pedestrian movement the contribution of NBS-led greening to liveliness conditions is particularly significant. A highway edge without tree cover is functionally inaccessible as public space for much of the year. The same edge, once planted with a continuous street tree corridor and ground-level shrub planting, becomes a shaded linear amenity that connects residential populations to transit stops, markets, and public facilities directly addressing the POI accessibility and walkability deficits recorded in low-liveliness wards (Karacor and Akcam, 2016; Jacobs, 1961).

The systemic connection between individually small NBS interventions is also important here. No single highway edge planting scheme, considered in isolation, can sufficiently alter the per capita green space figures of a ward. However, when NBS interventions are deployed at corridor scale as a connected network of street tree plantings, vegetated medians, under-flyover greening, and roadside parkways across the five major highway corridors, the cumulative contribution to urban tree canopy, biodiversity habitat, and pedestrian green space access becomes ecologically and socially significant (Forman et al., 2003; McPhearson et al., 2024; Frantzeskaki, 2019). The Priority Intervention Corridor framework developed in this study is designed precisely to enable this corridor-scale thinking by identifying not isolated sites but spatially contiguous ward clusters where a connected greening strategy can be planned and implemented.

6.3. NBS typology and site-specific suitability criteria

The identification of Priority Intervention Corridors at ward level is an essential first step in NBS planning, but it is not enough for implementation without site-specific translation. The NBS priority points included in the spatial overlay indicate locations where highway edge conditions have been assessed as suitable for NBS intervention based on measurable site-level criteria such as edge width, surface type, physical accessibility, and proximity to residential populations. These criteria are not applied uniformly across all edge types: the NBS interventions suitable for a wide, publicly accessible median strip are very different from those for a narrow road shoulder or an under-flyover space.

Specifically, highway edge spaces are suitable for tree planting when a minimum unobstructed width of 3 metres is available, the surface condition allows for soil preparation or the installation of raised planting beds, and the edge is physically accessible from the adjacent footpath or pedestrian zone. Vegetated buffer zones and shrub corridors are viable in narrower edges of 1.5–3 metres where structural tree planting is not feasible. Under-flyover spaces, with vertical clearance exceeding 4 metres and surface permeability suitable for planting, are deemed appropriate for ground-level greening such as grass cover, low shrub planting, and community gardens (Chatzimentor et al., 2020; Raymond et al., 2017; Kabisch et al., 2017). These thresholds, based on the highway edge typology mapping conducted in this study, offer a transparent, measurable, and adaptable basis for NBS suitability assessment in other city contexts.

This site-specific framing is significant because it directly addresses a limitation identified in the broader NBS literature that NBS intervention recommendations are frequently presented at a qualitative level without the measurable criteria necessary for planning departments to assess feasibility at the site scale (Grabowski et al., 2023; Frantzeskaki, 2019). By grounding the Priority Intervention Corridor framework in a typology-based suitability assessment with explicit width, surface, and accessibility thresholds, the method developed here provides a more actionable planning output than ward-level prioritisation alone.

6.4. Limitations of the framework

Six limitations of the framework are identified and discussed below. Table 6 provides a consolidated overview of each limitation, its source, its implication for the results reported in this study, and the refinement direction proposed for future applications.

Collectively, these six limitations do not undermine the principal finding of the study - that 82 wards across the GCC boundary meet the dual criteria of high green space deficit and low liveliness potential while containing confirmed NBS-suitable highway edge locations - but they qualify the precision with which individual ward scores and site-level NBS-SI values should be interpreted. The Priority Intervention Corridor designations are best understood as planning-scale indicators of where detailed site investigation is warranted, rather than as direct substitutes for site-level survey and engineering assessment.

6.5. Transferability to other Global South Cities

The framework developed in this study is designed to be transferable to other rapidly urbanising cities in the Global South that share Chennai's combination of high population density, severe green space deficit, inadequate park provision, and extensive highway infrastructure. The input data requirements are deliberately constrained to sources that are available without cost and at city scale in most urban contexts OpenStreetMap, administrative ward boundaries, census population data, and satellite imagery for highway edge digitisation. No remote sensing processing, proprietary datasets, or specialist software beyond an open-source GIS platform are required.

Cities sharing the structural conditions represented in Chennai including Colombo, Dhaka, Nairobi, Karachi, and Ho Chi Minh City face comparable challenges of green space deficit, rapid road network expansion, and limited land availability for conventional park provision (Aghaloo and Sharifi, 2024; Chatzimentor et al., 2020). In each of these

Table 6
Framework limitations: source, implication for results, and future refinement directions.

	Limitation	Source	Implication for Results	Future Refinement Direction
L1	**OSM green space underrepresentation**	OSM contributor density in Indian cities is lower than in European contexts. Private green space, institutional gardens, and informal vegetation are systematically unmapped (Lahoti et al., 2019).	Per capita green space values - including extreme values as low as 0.0005 m^2 - represent conservative estimates of formal mapped provision, not total urban vegetation. Deficit may be overstated in wards with significant unmapped private green cover.	Supplement OSM with NDVI-derived vegetation cover from Sentinel-2 or equivalent to capture informal and private green space independently of contributor mapping density.
L2	**Extreme per capita green space values as data artifacts**	OSM mapping precision limitations produce near-zero green space values in wards where formal parks are absent but informal vegetation exists at ground level.	Values as low as 0.0005 m^2 per capita reflect OSM data coverage gaps, not actual absence of all vegetation. All 200 wards fall below the 9 m^2 UN threshold regardless - the direction of the finding is robust even if precise values are conservative.	Cross-validate OSM-derived per capita figures against NDVI-based canopy cover estimates. Report both formal and informal green cover separately.
L3	**Cross-sectional analysis - no temporal dynamics**	Framework captures conditions at a single point in time (2025). Highway edge spaces are generated and modified continuously through ongoing infrastructure development. NBS interventions require longitudinal evaluation.	Priority Intervention Corridor designations reflect 2025 conditions. Wards near active road infrastructure projects may see edge space conditions change after analysis date. No phasing or staging of interventions over time is modelled.	Incorporate phased intervention scheduling based on projected infrastructure timelines. Add longitudinal monitoring indicators - NDVI change detection and pedestrian activity datasets - to evaluate NBS outcomes over time.
L4	**GIS overestimation of physical accessibility**	Satellite imagery at 624 m eye altitude cannot detect low-height barriers - compound walls, crash barriers, informal railings. GIS buffer analysis assumes access based on spatial proximity alone.	Field verification across 5 sites found GIS overestimated access in 4/5 cases. NBS-SI scores for corridors classified as Moderate or Limited Access may overstate immediate intervention feasibility. Access score is the framework's weakest parameter.	Conduct field verification of access conditions for all Tier 1 and Tier 2 corridors before NBS implementation planning. Integrate street-level imagery (Google Street View or equivalent) into accessibility assessment to reduce overestimation without full field surveys.
L5	**OSM POI coverage gaps and liveliness index incompleteness**	OSM POI data systematically underrepresents informal commercial activity in Indian cities. 12 wards record zero POI counts and are excluded from the composite index.	Liveliness potential is underestimated in wards with active informal economies. 12 excluded wards cannot be ranked relative to included wards. The highway edge proximity variable does not account for physical barriers confirmed by field verification.	Supplement OSM POIs with mobile activity data or field survey counts of informal commercial activity. Validate liveliness index using street-level pedestrian count data in a subset of wards.
L6	**Ward-level spatial aggregation (MAUP) and 2011 Census data**	GCC ward boundaries vary substantially in area. Aggregate per capita and density values are subject to the Modifiable Areal Unit Problem (Openshaw, 1984). Population data is from the 2011 Census - no ward-level projection was available from GCC.	Per capita and density figures are not directly comparable across wards of different sizes despite area normalisation. Population redistribution since 2011 may affect relative ward rankings, though broad spatial pattern of Priority Intervention Corridor concentration is unlikely to change substantially.	Apply framework at finer spatial resolution - census enumeration block level — when sub-ward population data becomes available. Update analysis with 2031 Census data upon release.

contexts, the highway edge typology mapping, deficit assessment, liveliness index construction, and spatial overlay approach developed here could be applied with locally calibrated thresholds adjusting the per capita green space reference standard to the relevant national planning benchmark, and adapting the NBS suitability criteria to local climate, road design standards, and maintenance capacity.

The framework is not proposed as a universal model but as a replicable spatial intelligence tool one that enables planning departments with limited technical capacity to identify, prioritise, and communicate NBS intervention opportunities across a city's highway network using open data and standard GIS methods. This positioning responds directly to the need identified in the literature for GIS-based NBS frameworks that are accessible to practitioners in low- and middle-income country contexts, rather than dependent on the data infrastructure and technical resources available only in high-income city administrations (Peiris, 2024;Sathyakumar et al., 2020; Lahoti et al., 2019).

7. Conclusion

This study presents a typology-based NBS suitability framework for identifying, classifying, and prioritising highway edge spaces as urban green infrastructure corridors in rapidly urbanising cities of the Global South, demonstrated through the Greater Chennai Corporation boundary. The principal finding is that 82 of 188 assessed GCC wards meet the dual criteria of high green space deficit and low urban liveliness potential, while containing confirmed NBS-suitable highway edge locations constituting Priority Intervention Corridors concentrated along Chennai's major highway alignments. These corridors are not randomly distributed: they are spatially concentrated in the northern industrial zone, the eastern coastal residential expansion corridor, the western mid-city highway alignments, and the southern GST Road zone precisely the populations most underserved by existing green infrastructure and most directly exposed to highway environmental burdens. The spatial correspondence between ecological deprivation and liveliness deficit across these wards confirms that highway edge reclamation through NBS is not merely a greening opportunity but a necessary urban regeneration strategy for communities where accessible vegetation, shaded public space, and ecological connectivity are simultaneously absent.

The methodological contribution of the study is fourfold. First, a ward-level green space deficit assessment quantifies per capita provision against national and international planning standards using open-source OSM data and standard GIS methods, producing results that are directly comparable across planning contexts. Second, a composite Urban Liveliness Potential Index constructed from POI density, population density, and highway edge proximity spatially identifies where NBS-led greening would generate the greatest liveliness co-benefit for underserved populations. Third, a five-typology NBS Suitability Index (NBS-SI), derived from four sub-classification dimensionsedge width, physical accessibility, ecological condition (NDVI), and urban context assigns each highway edge space a composite score ranging from 3 to 9, enabling transparent and replicable prioritisation at city scale. The 82 Priority Intervention Corridors are stratified into Tier 1 (NBS-SI 7–9: immediate intervention potential) and Tier 2 (NBS-SI 5–6: preparatory measures

required), providing planning departments with an operationally sequenced implementation framework rather than an undifferentiated priority list. Fourth, field verification across five representative sites one per typology confirms that the width-based typology score and NBS function assignment are reliable from GIS data alone, while identifying GIS overestimation of physical accessibility as the framework's primary refinement need, addressable through street-level imagery integration or targeted field checking of Tier 1 and Tier 2 corridors.

The framework's transferability to other Global South cities is a deliberate design feature. All required inputs OpenStreetMap data, administrative ward boundaries, census population figures, satellite imagery for NDVI extraction, and Google Earth Pro imagery for edge space digitisation are available at no cost in most rapidly urbanising contexts across South Asia, Sub-Saharan Africa, and Southeast Asia. The four-dimension NBS-SI scoring logic and five typology categories are not anchored to Indian national standards: planners in cities such as Colombo, Dhaka, Nairobi, Karachi, or Ho Chi Minh City can recalibrate width thresholds to local road reserve standards, NDVI thresholds to local vegetation density norms, and context categories to local planning priorities without altering the underlying classification structure. Three directions for future research are identified: supplementing OSM green space data with NDVI-derived canopy cover to reduce underrepresentation of informal vegetation; validating the liveliness potential index against street-level pedestrian activity data in a subset of wards; and establishing longitudinal monitoring indicators to evaluate NBS outcomes over time as Priority Intervention Corridor interventions are implemented.

CRediT authorship contribution statement

Gokul Kailash: Data curation, Formal analysis, Investigation, Methodology, Software, Writing – original draft. **Banu Chitra M:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

Aghaloo, K., Sharifi, A., Habibzadeh, N., Ali, T., Ru Chiu, Y., 2024. How nature-based solutions can enhance urban resilience to flooding and climate change and provide other co-benefits: a systematic review and taxonomy. Urban For. Urban Green. 95, 128320. https://doi.org/10.1016/j.ufug.2024.128320. September 2023.

Airbus, Pléiades Neo satellite imagery (Image acquisition date: 15 July 2026) [Satellite imagery], https://oneatlas.airbus.com, 2026, from Accessed via Airbus OneAtlas.

Chatzimentor, A., Apostolopoulou, E., Mazaris, A.D., 2020. A review of green infrastructure research in Europe: Challenges and opportunities. Landsc. Urban Plann. 198, 103775. https://doi.org/10.1016/j.landurbplan.2020.103775.

Choi, H.S.(Sunny), 2017. New forms of place-making and public space in contemporary urban development in Seoul, South Korea. Curr. Urban Stud. 05 (03), 332–347. https://doi.org/10.4236/cus.2017.53018. https://www.scirp.org/journal/paperinformation?paperid=78863.

Chennai Metropolitan Development Authority, 2008. Second Master Plan for Chennai Metropolitan Area, 2026. Chennai: CMDA.

Cohen-Shacham, E., Walters, G., Janzen, C., Maginnis, S., 2016. Nature-Based Solutions to Address Global Societal Challenges. IUCN, Gland, Switzerland. Xiii + 97pp.

Forman, R.T.T., Sperling, D., Bissonette, J.A., Clevenger, A.P., Cutshall, C.D., Dale, V.H., Winter, T.C., 2003. Road ecology: Science and solutions. Press, Washington, DC: Island.

Frantzeskaki, N., 2019. Seven lessons for planning nature-based solutions in cities. Environ. Sci. Policy 93, 101–111. https://doi.org/10.1016/j.envsci.2018.12.033 (December 2018).

Gómez-Varo, I., Miralles-Guasch, C., Marquet, O., 2026. When more isn't better: reevaluating urban vitality's role in fostering social cohesion. Cities 174, 106992. https://doi.org/10.1016/j.cities.2026.106992. https://linkinghub.elsevier.com/retrieve/pii/S0264275126002246.

Govindarajulu, D., 2014. Urban green space planning for climate adaptation in Indian Cities. Urban Clim. 10 (P1), 35–41. https://doi.org/10.1016/j.uclim.2014.09.006.

Grabowski, Z.J., McPhearson, T., Pickett, S.T.A., 2023. Transforming US urban green infrastructure planning to address equity. Landsc. Urban Plann. 229, 104591. https://doi.org/10.1016/j.landurbplan.2022.104591. https://linkinghub.elsevier.com/retrieve/pii/S0169204622002407.

Hamersma, M., Heinen, E., Tillema, T., Arts, J., 2017. The development of highway nuisance perception: Experiences of residents along the Southern Ring Road in Groningen, the Netherlands. Land Use Policy 61, 553–563. https://doi.org/10.1016/j.landusepol.2016.12.001.

Harnack, M., Kohler, M., 2011. As found: Use, meaning and re-appropriation of contentious urban spaces. In: Hauck, T., Keller, R., Kleinekort, V. (Eds.), Infrastructural urbanism: Addressing the in-between. DOM Publishers, Berlin, pp. 131–139.

Huang, L., Qiu, J., Wu, J., 2024. Promoting urban-rural landscape sustainability through geodesign. Landscape Ecol. 39 (10). https://doi.org/10.1007/s10980-024-01973-2.

Iungman, T., Cirach, M., Marando, F., Pereira Barboza, E., Khomenko, S., Masselot, P., Quijal-Zamorano, M., Mueller, N., Gasparrini, A., Urquiza, J., Heris, M., Thondoo, M., Nieuwenhuijsen, M., 2023. Cooling cities through urban green infrastructure: a health impact assessment of European cities. Lancet 401 (10376), 577–589. https://doi.org/10.1016/s0140-6736(22)02585-5. https://linkinghub.elsevier.com/retrieve/pii/S0140673622025855.

Jacobs, J., 1961. The death and life of great American cities. Random House, New York.

Kabisch, N., Korn, H., Stadler, J., Bonn, A., 2017. Nature-based solutions to climate change adaptation in urban areas—linkages between science, policy and practice. Theory Practice of Urban Sustainability Transitions. Springer International Publishing, Cham, ISBN 9783319537504, pp. 1–11. https://doi.org/10.1007/978-3-319-56091-5_1. http://link.springer.com/10.1007/978-3-319-56091-5_1. (Accessed 2 September 2017).

Kailash, G., Banu Chitra, M., 2025. Transforming urban neglected spaces: the role of adaptive reuse and nature-based solutions. Int. Res. J. Multidiscip. Scope 6 (2), 1027–1050. https://doi.org/10.47857/irjms.2025.v06i02.03699.

Kamalipour, H., Peimani, N., 2015. Assemblage thinking and the city: implications for urban studies. Curr. Urban Stud. 03 (04), 402–408. https://doi.org/10.4236/cus.2015.34031. https://www.scirp.org/journal/paperinformation?paperid=62067.

Kamalipour, H., Peimani, N., 2015. Assemblage thinking and the city: implications for urban studies. Curr. Urban Stud. 03 (04), 402–408. https://doi.org/10.4236/cus.2015.34031. https://www.scirp.org/journal/paperinformation?paperid=62067.

Kaźmierczak, Aleksandra, 2013. The contribution of local parks to neighbourhood social ties. Landscape and Urban Planning 109 (1), 31–44. https://doi.org/10.1016/j.landurbplan.2012.05.007. https://linkinghub.elsevier.com/retrieve/pii/S0169204612001569.

Karacor, E.K., Akcam, E., 2016. Comparative analysis of the quality perception in public spaces of duzce city. Curr. Urban Stud. 04 (03), 257–266. https://doi.org/10.4236/cus.2016.43017. https://www.scirp.org/journal/paperinformation?paperid=69698.

Lahoti, S., Lahoti, A., Saito, O., 2019. Benchmark assessment of recreational public Urban Green space provisions: a case of typical urbanizing Indian City, Nagpur. Urban Forest. Urban Green. 44, 126424. https://doi.org/10.1016/j.ufug.2019.126424. https://linkinghub.elsevier.com/retrieve/pii/S1618866719301062.

Lehmann, S., 2021. La naturaleza En El contexto urbano: La renaturalización como Una dimensión importante De La resistencia Y La planificación urbana. Módulo Arquit. - CUC 26 (26), 161–190. https://doi.org/10.17981/mod.arq.cuc.26.1.2021.07.

Li, M., Remme, R.P., van Bodegom, P.M., van Oudenhoven, A.P.E., 2025. How do nature-based solutions contribute to biodiversity in cities? Ecol. Indic. 175. https://doi.org/10.1016/j.ecolind.2025.113523. March.

Malaya, U., U. Malaya, U. Tun, and H. Onn. 2024. "Urban voids : nature-based solutions (NBS) TO." 22(3):425–41.

Masiero, M., Biasin, A., Amato, G., Malaggi, F., Pettenella, D., Nastasio, P., Anelli, S., 2022. Urban forests and green areas as nature-based solutions for brownfield redevelopment: a case study from brescia municipal area (Italy). Forests 13 (3), 1–28. https://doi.org/10.3390/f13030444.

Mastervich, B., Garbach, K., Harwell, M.C., 2024. Enhancing multiple benefits of brownfield cleanups by applying ecosystem services concepts. Front. Ecol. Evol. 12, 1–10. https://doi.org/10.3389/fevo.2024.1286150 (February).

McConnell, V., Walls, M., 2005. The value of open space: evidence from studies of nonmarket benefits. Resour. Future 82 (January).

McPhearson, T., Frantzeskaki, N., Ossola, A., Diep, L., Anderson, P., et al., 2025. Global synthesis and regional insights for mainstreaming urban nature-based solutions. Proceedings of the National Academy of Sciences 122, e2315910121. https://doi.org/10.1073/pnas.2315910121.

Mohamed Anuar, M.I.N., Abdullah, S.A., 2020. Reappropriation of elevated highway residual space through green infrastructure planning. Plann. Malaysia 18 (4), 203–219. https://doi.org/10.21837/pm.v18i14.827.

OpenStreetMap contributors, 2026. OpenStreetMap (Road network, buildings, land use, and water bodies). OpenStreetMap Foundation [Data set] Retrieved July 15, 2026, from. https://www.openstreetmap.org.

Peiris, M.T.O.V., 2024. Assessment of urban resilience to floods: A spatial planning framework for cities. Sustainability 16 (20), 9117. https://doi.org/10.3390/su16209117.

Raymond, C.M., Frantzeskaki, N., Kabisch, N., Berry, P., Breil, M., Nita, M.R., Geneletti, D., Calfapietra, C., 2017. A framework for assessing and implementing the co-benefits of nature-based solutions in urban areas. Environment. Sci. Pol. 77, 15–24. https://doi.org/10.1016/j.envsci.2017.07.008.

Sanches, P.M., Pellegrino, P.R.M., 2016. Greening potential of derelict and vacant lands in urban areas. Urban For. Urban Green. 19, 128–139. https://doi.org/10.1016/j.ufug.2016.07.002.

Sathyakumar, V., Ramsankaran, R., Bardhan, R., 2020. Geospatial approach for assessing spatiotemporal dynamics of urban green space distribution among neighbourhoods: A demonstration in Mumbai. Urban For. Urban Green. 48, 126585. https://doi.org/10.1016/j.ufug.2020.126585.

Le Texier, M., Schiel, K., Caruso, G., 2018. The provision of urban green space and its accessibility: spatial data effects in brussels. PLoS ONE 13 (10), 1–17. https://doi.org/10.1371/journal.pone.0204684.

Trancik, R., 1986. Finding lost space: Theories of urban design. Van Nostrand Reinhold, New York.

van den Berg, M., Wendel-Vos, W., van Poppel, M., Kemper, H., van Mechelen, W., Maas, J., 2015. Health benefits of green spaces in the living environment: a systematic review of epidemiological studies. Urban Fores. Urban Green. 14 (4), 806–816. https://doi.org/10.1016/j.ufug.2015.07.008. https://linkinghub.elsevier.com/retrieve/pii/S1618866715001016.

Department of Architecture, University of Moratuwa, Sri Lanka, Weththasinghe, A., Wijesundara, J., 2017. In: Reclaiming Traffic Influenced Urban Residual Spaces for The Public in Colombo [Journal-article], Sri Lanka.

Xue, B., Zhang, J., Liu, B., Alberti, M., Fan, Y., Zhao, D., Grimm, N.B., 2025. Accessibility–equality dynamics in urban blue and green spaces reshape human well-being [Data set]. Zenodo (CERN European Organization for Nuclear Research). https://doi.org/10.5281/zenodo.17411073.

Yang, B., Li, S., 2016. Design with nature: Ian McHarg's ecological wisdom as actionable and practical knowledge. Landsc. Urban Plann. 155, 21–32. https://doi.org/10.1016/j.landurbplan.2016.04.010. https://linkinghub.elsevier.com/retrieve/pii/S0169204616300482.